

Research Methodology and IPR (22MBA23)

Module- 01

Syllabus

Module-1

Introduction to Business Research:

Meaning, types, process of research- management problem, defining the research problem, formulating the research Hypothesis, developing the research proposals, research design formulation, sampling design, planning and collecting the data for research, data analysis and interpretation. Research Application in business decisions, Ethical issues in business research. Features of a good research study.

Research

Meaning:

- Research in common man's language refers to “search for knowledge”
- Research is an art of scientific investigation.
- The Advanced Learner's Dictionary of Current English lays down the meaning of research as,

a careful investigation or inquiry specially through search for new facts in any branch of knowledge”.

It is also a systematic design, collection, analysis and reporting the findings and solutions for problems of a company.

- Research is a structured enquiry that utilizes acceptable scientific methodology to solve problems and create new knowledge that is generally applicable.
- Research is the systematic process of collecting and analyzing information (data) in order to increase our understanding of the phenomenon about which we are concerned or interested.
- Research is an art of scientific investigation. It is also a systematic design, collection, analysis and reporting the findings and solutions for the problems of a company.

Definition

- Research can be defined as an organized and systematic study of materials and sources in order to discover new things and establish facts and reach new conclusions. Essentially speaking, research involves a well-developed plan, a systematic approach to develop new theories or finding solutions to various problems.
- According to Clifford Woody, “research comprises defining and refining problems, formulating hypothesis or suggested solutions, collecting, organizing and evaluating data, making decisions and reaching conclusions to determine whether they fit the formulating hypothesis”

Business Research

- A Systematic inquiry that provides information to guide managerial decisions.
- Systematic and objective process of gathering, recording and analyzing data for aid in making business decisions.

Scientific Method

Scientific research is one which yields the same results when repeated by different individuals. Techniques or procedures used to analyze empirical evidence in an attempt to confirm or disprove prior concepts

Empirical means verifiable by observation, experimentation or experience.

A scientific undertaking which by means of logical and systematic techniques, aims to:

1. Discover new facts or verify and test old facts.
2. Analyze their sequences, interrelationships and causal explanations.
3. Develop new scientific tools, concepts and theories, which would facilitate valid and reliable study of human behavior.

Characteristics of Scientific Method

- **Validity** - Validity is the ability of a measuring instrument to measure what it is supposed to.
- **Reliability** - Measurement by instruments such as thermometer can be repeated any number of times by different individuals, but the results will be the same.

Types of research

1. Basic research:
2. Applied Research
3. Qualitative Research
4. Quantitative Research
5. Historical Research
6. Exploratory Research:
7. Descriptive Research
8. Causal Research
9. Simulation research

1. Basic research:

research for the sake of enhancing knowledge is termed as basic research.

- It is also called as theoretical research
- Basic or pure or fundamental research is done with the intention of empowering the unknown, Undertaken for the sake of knowledge without any intention to apply or practice.
- It is primarily concerned with developing and formulating theories and generalization.
- It addresses itself to more fundamental questions and not the problems with immediate commercial potential.
- Not directly involved with practical problems.

Contributions:

- By developing principle, pure research *offers solutions* to many *practical problems*.
 - Helps to *find critical factors* in a practical problem.
 - Develops many *alternative solutions*.
 - How did the universe begin?
- What are protons, neutrons, and electrons composed of?
 - What is the specific genetic code of the fruit fly?
 - Example : Theory of Relativity (by Einstein)

2. Applied Research

- It is also termed as need based research having high practical relevance.
- Specific in nature, result oriented and is driven by clear aim.
- Undertaken to find solutions for real-life problem requiring an action or policy decision.
- It may incidentally contribute to the development of theoretical knowledge.

Contributions:

- Can contribute new facts about existing theories.
- Puts theories to test.
- May aid in conceptual clarification.
- Integrate previously existing theories. Example :
- To develop a new market for the product.
- improve the energy efficiency of homes, offices, or modes of transportation

3. Qualitative Research

An unstructured, exploratory research methodology based on small samples that provides insights and understanding of the problem setting.

- The aim is a complete, detailed description.
- Data is in the form of words, pictures or objects.
- Case studies, group interviews

4. Quantitative Research

A research methodology that seeks to quantify the data and typically applies some form of statistical analysis.

- The aim is to classify features, count them, and construct statistical models in an attempt to explain what is observed.
- Data is in the form of numbers and statistics.
- Experiments and surveys

5. Historical Research

- Historical study is a study of past records and data in order to understand the future trends and development of the organization or market.
- There is no direct observation.
- Research has to depend on the conclusions or inferences drawn in the past.
- Main objective is to derive explanation and generalization from the past trends in order to understand the present and anticipate the future.
- Example : Investors in the share market study the past records or prices of shares which he/she intends to buy. Studying the share prices of a particular company enables the
- investor to take decision whether to invest in the shares of a company.

6. Exploratory Research:

This kind of research is carried out at the very beginning when the problem is not clear or vague.

- Provides insights into and comprehension of an issue or situation.
- Often relies on secondary research such as reviewing available literature and/or data, or qualitative approaches such as informal discussions with consumers, employees,
- management or competitors and more formal approaches through in-depth interviews, focus groups, case studies or pilot studies.
- Example : Exploring the reasons for sales decline of a product in a company.

7. Descriptive Research

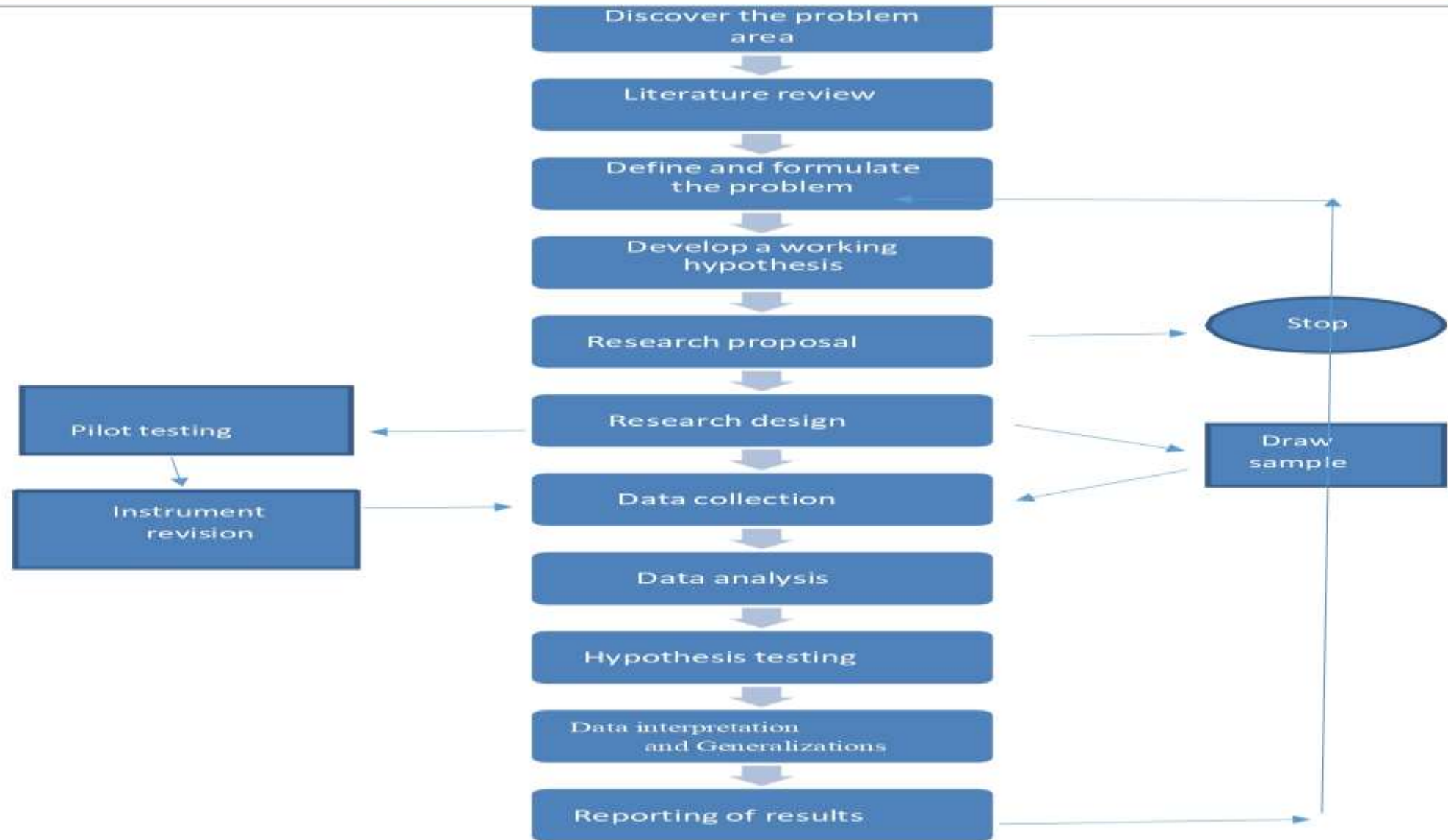
- Describes data and characteristics about the population or phenomenon being studied.
- Answers the questions who, what, where, when and how.
- Describes the characteristics of the respondent in relation to a particular product. Deals with demographic characteristics of the consumer.
- Example : Degree of viewing TV Channels, its variation with age, income level, profession of respondent as well as time of viewing.

8. Causal Research

- Conducted to determine the cause and effect relationship between the two variables.
- Way of seeing how actions now will affect a business in the future.
- Example:
 - □ Training and productivity of an employees
 - □ Advertisement and sales

9. Simulation research

- This research involves the creation of an artificial environment which is very similar to the real environment. Therefore with in the artificial environment the variables are manipulated and studied.
- Ex: study of consumer buying behavior by giving specific amount of time and money in a mall.



Management problem

- When a problem has been clearly and accurately identified can a research project be conducted properly. It is a concise description of the issues that need to be addressed by a problem solving team and should be presented to them (or created by them) before they try to solve the problem. It asks what the decision maker needs to do. Mostly action oriented and Focuses on symptoms.

Defining the research problem

- The research problem is a general statement of an issue meriting research. Its nature will suggest appropriate forms for its investigation
- Problem definition involves stating the general marketing research problem and identifying its specific components. Only when research problem has been clearly defined can research be designed and conducted properly.

Basis of comparison	Management problem	Research problem
Nature	Asks what the Decision Makers needs to do	Asks what information needed and how it can be obtained
Orientation	Action Oriented	Information oriented
Focus	Focuses on Symptoms	Focuses on underlying causes
Common errors	Problem definition is too broad	Problem definition is too narrow
example	Should a new product be introduced	To determine consumer preferences and purchase intentions for the proposed new product
example	Should the advertising campaign be changed	To determine the effectiveness of the current advertising campaign
example	Should the price of the brand be increased	To determine the price elasticity of the demand and the impact on sales and profits of various levels of price changes

Formulating the research Hypothesis

A good hypothesis relates and explains the known facts. It should also predict new facts. It *must* be stated in such a way that we can test it by experimentation or further observation, or it is of no scientific value. Also, it must be stated in a way that would allow us to show if it is incorrect, i.e., it must be "falsifiable." A scientist *must* be willing to accept the possibility that his or her hypothesis is incorrect, and this point often separates true science from pseudoscience. (In fact, most scientists work hard to develop good hypotheses, and then spend a great deal of effort trying to *disprove* them. Pseudo scientists tend to settle on a hypothesis that suits their needs or expectations, and then spend a great deal of effort trying to prove that they are "*true*".

Developing the research proposals

Research proposal is outline, a draft plan of a research work proposed by the researcher. It is also termed as synopsis. It contain

- A statement of the problem
- The proposed methodology
- The benefits of the research
- The reasons required to conduct the research

Contents of research proposal

The research proposal contents differ depending on the type of proposal and intended audience.

- Problem statement
- Research objectives
- Literature review
- Importance/ benefits of study
- Research design
- Data analysis
- Nature and form of results
- Schedule
- Facilities and special resources
- Bibliography
- Appendices
- Measurement instrument

Research design formulation

- A research design is a framework or blueprint for conducting the marketing research project. It details the procedures necessary for obtaining the required information, and its purpose is to design a study that will test the hypotheses of interest, determine possible answers to the research questions, and provide the information needed for decision making. Decisions are also made regarding what data should be obtained from the respondents (e.g., by conducting a survey or an experiment). A questionnaire and sampling plan also are designed in order to select the most appropriate respondents for the study.
- Secondary data analysis (based on secondary research)
- Qualitative research
- Methods of collecting quantitative data (survey, observation, and experimentation)
- Definition of the information needed
- Measurement and scaling procedures
- Questionnaire design
- Sampling process and sample size
- Plan of data analysis

The following steps are involved in formulating a research design:

Sampling design

- Sampling is a means of selecting a subset of units from a target population for the purpose of collecting information. This information is used to draw inferences about the population as a whole. The subset of units that are selected is called a sample. The sample design encompasses all aspects of how to group units on the frame, determine the sample size, allocate the sample to the various classifications of frame units, and finally, select the sample. Choices in sample design are influenced by many factors, including the desired level of precision and detail of the information to be produced, the availability of appropriate sampling frames, the availability of suitable auxiliary
- variables for stratification and sample election, the estimation methods that will be used and the available budget in terms of time and resources.

Data analysis and interpretation

- Analysis of data is a process of inspecting, cleaning, transforming, and modeling data with the goal of discovering useful information, suggesting conclusions, and supporting decision-making. Data analysis has multiple facets and approaches, encompassing diverse techniques under a variety of names, in different business, science, and social science domains.

Literature review

It involves a comprehensive review of published and unpublished work from the secondary sources of data available in the relevant area of study.

Importance of literature review:

- ❖ Identification of problem for the study
- ❖ Monitors the repetition
- ❖ Helps avoiding mistakes
- ❖ Creates appropriate theoretical structure
- ❖ Facilitates interpretation of study

Planning and collecting the data for research

- Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes. Generally there are three types of data collection and they are 1. Surveys: Standardized paper-and-pencil or phone questionnaires that ask predetermined questions.
- Interviews: Structured or unstructured one-on-one directed conversations with key individuals or leaders in a community.
- Focus groups: Structured interviews with small groups of like individuals using standardized questions, follow-up questions, and exploration of other topics that arise to better understand participants

Research application in business decision

Businesses use research to ascertain the success of their advertising. For instance, a dairy manufacturer may want to find out what percentage of the people saw its latest TV commercial. The dairy manufacturer may find that more individuals become aware of its advertising the longer the TV ad runs. The company may decide to run its TV ads at different times if few people have seen the ads.

A business can make well-informed decisions due to research. In the research process, the business will be able to acquire details about key business areas, analyze it, create a strategy and distribute business information. Reports, presented to the top management, often contain details on consumer and employee preferences and all the available channels for sales, marketing, finance and production. Management makes use of these details to **determine the best strategy.**

Research is a necessity at all stages and phases of business operations. Initial research is needed to evaluate whether getting into the given kind of business would be profitable and whether there exists demand for the proposed product

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Regarding the staff, a correctly carried out research can uncover important details on their satisfaction quotient, the difficulties experienced by them and how the problems related to relationships at the workplace could be handled. An analysis into the results would allow the management to bring about changes for the all round effective functioning of the organization and its employees. **The workers can be trained and coached in line with the needs.** This would help personal as well as professional development improving overall organizational performance.

Research is important for **managerial decision making**. All strategic business areas are analyzed and evaluated; then techniques for more efficient procedures are created. Through proper research, the organization will be able to pick the most effective, productive and profitable one.

Research could possibly be applied to marketing, production, finance, IT and Human resources.

Research can answer questions for **various problems**, from getting a grip on industry trends, identifying new products to produce and deliver to the market, or deciding on which site to locate an outlet, to better understanding what it needs to fulfill customer demands. Research can also help evaluate if a product is accepted in the market. Research aids expansion into new markets

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Research helps in **testing the potential success of new products**. Businesses have to understand what kinds of products consumers would like before they market them. For instance, a restaurant may in the beginning, interview focus groups to test types of burgers. The focus groups will probably include small teams of consumers. The goal of the focus group may be to figure out which burger customers prefer. Ultimately, the company may test the burgers through surveys with larger groups of people. The above points state the importance of research in business decision making. Research is necessary to gather facts and statistics with regards to a company's customers, employees and competitors. Based on these numbers, businesses are able to make better managerial decisions.

New Knowledge: The fascination and desire for new knowledge, new facts for business cycles, environment analysis and technological up gradation are the primary reasons of research.

Research assists the company to find the right supplier at the right price and at the right time. An appropriate supplier choice makes it possible for the company to obtain or acquire top quality raw materials which result into production of good quality items which are consumed by the end user.

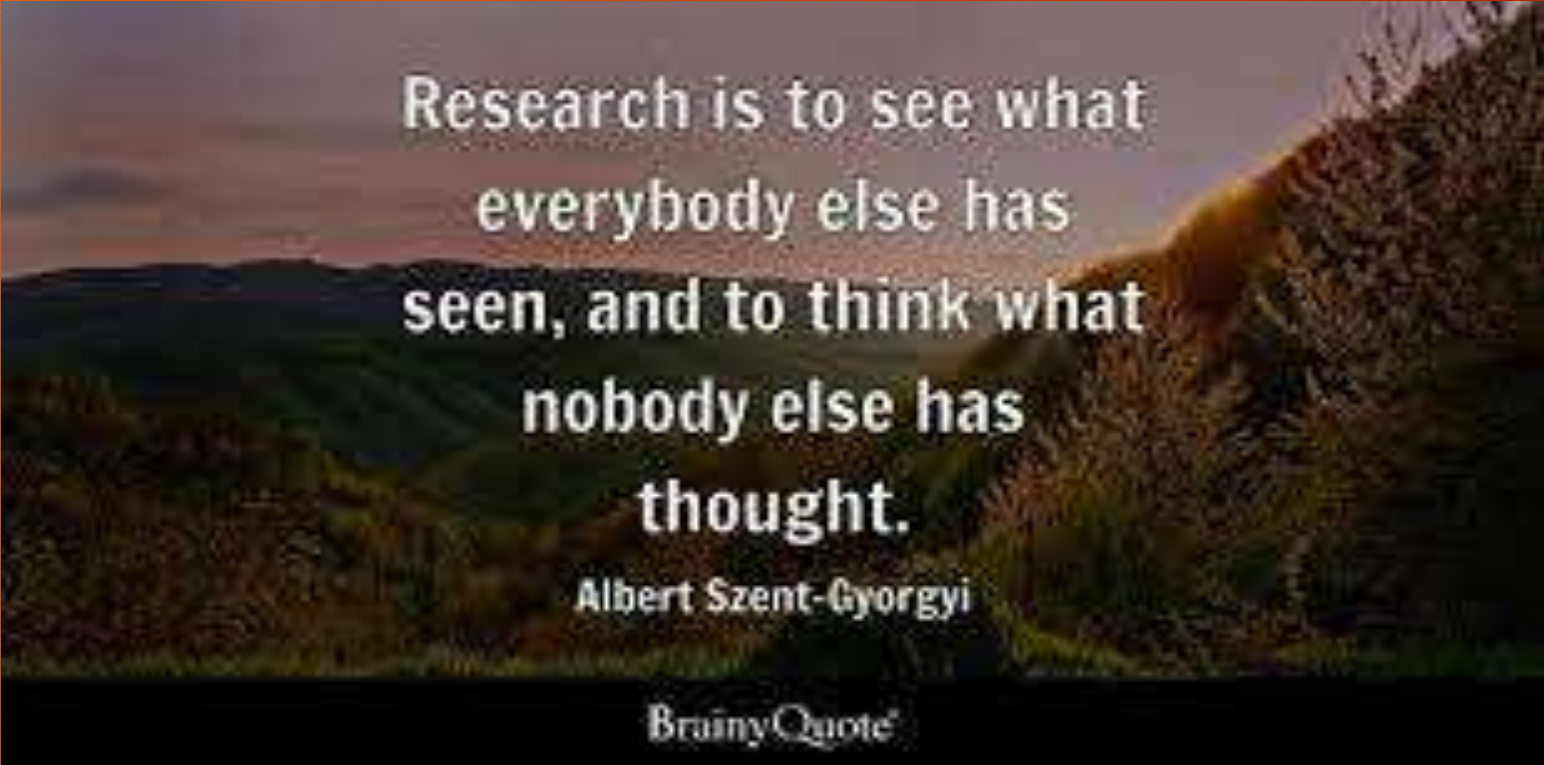
Ethical issues in business research.

- 1 . Study design and ethics approval
2. Data analysis
3. Authorship
4. Conflicts of interest
5. Redundant publication and plagiarism

Features of a good research study.

- ☐ Systematic
- ☐ Logical
- ☐ Empirical
- ☐ Replicable
- ☐ Purpose clearly defined
- ☐ Research process detailed
- ☐ Research design thoroughly planned
- ☐ High ethical standards applied
- ☐ Limitations frankly revealed
- ☐ Analysis adequate for decision maker's needs
- ☐ Findings presented unambiguously
- ☐ Conclusions justified

<<<<<<<THANK YOU>>>>>>



Research is to see what
everybody else has
seen, and to think what
nobody else has
thought.

Albert Szent-Gyorgyi

BraünyQuote®

Business Research Design:

MODULE-2

Syllabus

- ▶ **Business Research Design:**
- ▶ Meaning, types and significance of research design, errors affecting research design.
- ▶ **Exploratory Research:** Meaning, purpose, methods, Literature search, experience survey, focus groups and comprehensive case methods.
- ▶ **Conclusive Research Design:** Descriptive Research, Meaning, Types, Cross sectional studies and longitudinal studies.
- ▶ **Experimental Research Design:** Meaning and classification of experimental designs, formal and informal, Pre experimental design, True experimental design, Quasi-experimental design, Statistical experimental design.

Business Research Design

Meaning

- ▶ The formidable problem that follows the task of defining the research problem is the preparation of the design of the research project, popularly known as the “research design”.
- ▶ Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study constitute a research design. **“A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure.”**¹ in fact, the research design is the conceptual structure within which research is conducted; it constitutes the blueprint for the collection, measurement and analysis of data. As such the design includes an outline of what the researcher will do from writing the hypothesis and its operational implications to the final analysis of data. More explicitly, the designing decisions happen to be in respect of:
 - ▶ What is the study about?
 - ▶ Why is the study being made?
 - ▶ Where will the study be carried out?
 - ▶ What type of data is required?
 - ▶ Where can the required data be found?
 - ▶ What periods of time will the study include?
 - ▶ What will be the sample design?
 - ▶ What techniques of data collection will be used?
 - ▶ How will the data be analysed?

- ▶ In what style will the report be prepared?
- ▶ Keeping in view the above stated design decisions, one may split the overall research design into the following parts:
 - ▶ *The sampling design* which deals with the method of selecting items to be observed for the given study;
 - ▶ *The observational design* which relates to the conditions under which the observations are to be made;
 - ▶ *The statistical design* which concerns with the question of how many items are to be observed and how the information and data gathered are to be analysed; and
 - ▶ *The operational design* which deals with the techniques by which the procedures specified in the sampling, statistical and observational designs can be carried out.
- ▶ From what has been stated above, we can state the important features of a research design as under:
 - ▶ It is a plan that specifies the sources and types of information relevant to the research problem.
 - ▶ It is a strategy specifying which approach will be used for gathering and analysing the data.
 - ▶ It also includes the time and cost budgets since most studies are done under these two constraints.
 - ▶ In brief, research design must, at least, contain—(a) a clear statement of the research problem;
 - ▶ Procedures and techniques to be used for gathering information; (c) the population to be studied; and (d) methods to be used in processing and analysing data.

Types of Research Design

There are different types of research design depend on the nature of the problem and objectives of the study. Following are the four types of research design.

- ▶ Explanatory Research Design
- ▶ Descriptive Research Design
- ▶ Diagnostic Research Design
- ▶ Experimental Research Design

1. Explanatory Research Design

- ▶ In explanatory research design a researcher uses his own imaginations and ideas. It is based on the researcher personal judgment and obtaining information about something. He is looking for the unexplored situation and brings it to the eyes of the people. In this type of research there is no need of hypothesis formulation.

2. Descriptive Research Design

- ▶ In descriptive research design a researcher is interested in describing a particular situation or phenomena under his study. It is a theoretical type of researcher design based on the collection designing and presentation of the collected data. Descriptive research design covers the characteristics of people, materials, Scio-economics characteristics such as their age, education, marital status and income etc. The qualitative nature data is mostly collected like knowledge, attitude, beliefs and opinion of the people. Examples of such designs are the newspaper articles, films, dramas, and documentary etc.

3. Diagnostic Research Design

Here researcher wants to know about the root causes of the problem. He describes the factors responsible for the problematic situation. It is a problem solving research design that consists mainly:

- ▶ Emergence of the problem
- ▶ Diagnosis of the problem
- ▶ Solution for the problem and
- ▶ Suggestion for the problem solution

4. Experimental Research Design

In this type of research design is often uses in natural science but it is different in social sciences. Human behavior cannot be measured through test-tubes and microscopes. The social researcher use a method of experiment in that type of research design. One group is subjected to experiment called independent variables while other is considered as control group called dependent variable. The result obtained by the comparison of both the two groups. Both have the cause and effect relationship between each other.

Significance of Research Design

- ▶ Research design offers the investigator an opportunity to carry out different research operations efficiently. This makes research as valuable as possible producing maximum information with minimum effort, time and money.
- ▶ Researcher needs to consider all necessary precautions when preparing the design, as any error may upset the whole project. The reliability of result, which a researcher is looking, is proportional with design that constitutes a firm foundation of entire body of research work.

Significance of Research Design in Research Methodology

- ▶ *Research design is significant* simply because it allows for the smooth sailing of the various research operations, thus making research as efficient as possible producing maximum information with nominal expenses of effort, time and money.
- ▶ Just as for better, economical and attractive construction of a home, we require a blueprint (or what is typically known as the map of the home) well planned and prepared by an expert architect, in the same way we require a design or a plan in advance of data collection and analysis for our research study. It means advance planning of the techniques to be implemented for accumulating the appropriate data and the strategies to be employed in their analysis, keeping in view the purpose of the research and the availability of staff, time and money.
- ▶ Preparation of the design must be carried out meticulously as any error in it may upset the complete project. Research design, actually, has a great significance and impact on the reliability of the results achieved and as such constitutes the firm base of the entire edifice of the research work.
- ▶ Even then the necessity for a well planned design is at times not realized by many people. The significance which this problem warrants is not given to it. Because of this many researches do not serve the purpose for which they are undertaken. The truth is, they may even provide misleading conclusions.
- ▶ Thoughtlessness in developing the research project may lead to rendering the research exercise futile. It is, for that reason, crucial that an efficient and appropriate design should be prepared before beginning research operations.
- ▶ The design assists the researcher to organize his ideas in a form whereby it will be possible for him to watch out for flaws and inadequacies. This type of design can also be given to others for their comments and critical evaluation. In the absence of such a strategy, it will likely be challenging for the critic to supply a comprehensive review of the offered study.
- ▶ A research design isn't just a work plan. A work plan details what needs to be done to complete the project but the work plan will flow from the project's design. The **function of a research design** is to make certain that evidence obtained allows us to answer the initial question as unambiguously as possible. Acquiring relevant evidence involves specifying the kind of evidence required to answer the research question, to test a theory, to judge a programme or to precisely describe some phenomenon.

Errors Affecting Research Designs

The research design must attempt to reduce the 8 types of potential errors that can influence research results, viz.

1. Surrogate Information error
2. Measurement error.
3. Experimental error.
4. Population Specification error.
5. Frame Error.
6. Sampling error.
7. Selection error.
8. Non Response error.

Exploratory research: Definition

- ▶ Exploratory research is defined as a research used to investigate a problem which is not clearly defined. It is conducted to have a better understanding of the existing research problem, but will not provide conclusive results. For such a research, a researcher starts with a general idea and uses this research as a medium to identify issues, that can be the focus for future research. An important aspect here is that the researcher should be willing to change his/her direction subject to the revelation of new data or insight. Such a research is usually carried out when the problem is at a preliminary stage. It is often referred to as grounded theory approach or interpretive research as it used to answer questions like what, why and how.

Types and methodologies of Exploratory research



Exploratory research: Steps to conduct a research

- ▶ **Identify the problem:** A researcher identifies the subject of research and the problem is addressed by carrying out multiple methods to answer the questions.
- ▶ **Create the hypothesis:** When the researcher has found out that there are no prior studies and the problem is not precisely resolved, the researcher will create a hypothesis based on the questions obtained while identifying the problem.
- ▶ **Further research:** Once the data has been obtained, the researcher will continue his study through descriptive investigation. Qualitative methods are used to further study the subject in detail and find out if the information is true or not.

Characteristics of Exploratory research

- ❖ They are not structured studies
- ❖ It is usually low cost, interactive and open ended.
- ❖ It will enable a researcher answer questions like what is the problem? What is the purpose of the study? And what topics could be studied?
- ❖ To carry out exploratory research, generally there is no prior research done or the existing ones do not answer the problem precisely enough.
- ❖ It is a time consuming research and it needs patience and has risks associated with it.
- ❖ The researcher will have to go through all the information available for the particular study he is doing.
- ❖ There are no set of rules to carry out the research per se, as they are flexible, broad and scattered.
- ❖ The research needs to have importance or value. If the problem is not important in the industry the research carried out is ineffective.
- ❖ The research should also have a few theories which can support its findings as that will make it easier for the researcher to assess it and move ahead in his study
- ❖ Such a research usually produces qualitative data, however in certain cases quantitative data can be generalized for a larger sample through use of surveys and experiments.

Advantages of Exploratory research

- ▶ The researcher has a lot of flexibility and can adapt to changes as the research progresses.
- ▶ It is usually low cost.
- ▶ It helps lay the foundation of a research, which can lead to further research.
- ▶ It enables the researcher understand at an early stage, if the topic is worth investing the time and resources and if it is worth pursuing.
- ▶ It can assist other researchers to find out possible causes for the problem, which can be further studied in detail to find out, which of them is the most likely cause for the problem.

Disadvantages of Exploratory research

- ▶ Even though it can point you in the right direction towards what is the answer, it is usually inconclusive.
- ▶ The main disadvantage of exploratory research is that they provide qualitative data. Interpretation of such information can be judgmental and biased.
- ▶ Most of the times, exploratory research involves a smaller sample, hence the results cannot be accurately interpreted for a generalized population.
- ▶ Many a times, if the data is being collected through secondary research, then there is a chance of that data being old and is not updated.

Importance of Exploratory research

- ▶ Exploratory research is carried out when a topic needs to be understood in depth, especially if it hasn't been done before. The goal of such a research is to explore the problem and around it and not actually derive a conclusion from it. Such kind of research will enable a researcher to set a strong foundation for exploring his ideas, choosing the right research design and finding variables that actually are important for the analysis. Most importantly, such a research can help organizations or researchers save up a lot of time and resources, as it will enable the researcher to know if it worth pursuing.

Conclusive Research Design

- ▶ Conclusive research design, as the name implies, is applied to generate findings that are practically useful in reaching conclusions or decision-making. In this type of studies research objectives and data requirements need to be clearly defined. Findings of conclusive studies usually have specific uses. Conclusive research design provides a way to verify and quantify findings of exploratory studies.
- ▶ Conclusive **research** design usually involves the application of quantitative methods of data collection and data analysis. Moreover, conclusive studies tend to be deductive in nature and research objectives in these types of studies are achieved via testing hypotheses.

Factor	Conclusive	Exploratory
Objectives	To test hypothesis and relationships	To get insights and understanding
Characteristics	Information needs a clearly defined Research process is formal and structured Large representative sample Data analysis is quantitative	Information needs are loosely defined Research process is unstructured and flexible Small, non-representative sample Primary data analysis is qualitative
Findings	Conclusive	Only tentative
Outcome	Findings used as input to decision making	Generally followed by further exploratory conclusive research

Descriptive Research -Meaning

- ▶ The descriptive research design involves observing and collecting data on a given topic without attempting to infer cause-and-effect relationships. The goal of descriptive research is to provide a comprehensive and accurate picture of the population or phenomenon being studied and to describe the relationships, patterns, and trends that exist within the data.
- ▶ Descriptive research methods can include surveys, observational studies, and case studies, and the data collected can be qualitative or quantitative. The findings from descriptive research provide valuable insights and inform future research, but do not establish cause-and-effect relationships.

Types of Descriptive Research



▶ 1. Survey Research

- ▶ Surveys are a type of descriptive research that involves collecting data through self-administered or interviewer-administered questionnaires. Additionally, they can be administered in-person, by mail, or online, and can collect both qualitative and quantitative data.

▶ 2. Observational Research

- ▶ Observational research involves observing and collecting data on a particular population or phenomenon without manipulating variables or controlling conditions. It can be conducted in naturalistic settings or controlled laboratory settings.

▶ 3. Case Study Research

- ▶ Case study research is a type of descriptive research that focuses on a single individual, group, or event. It involves collecting detailed information on the subject through a variety of methods, including interviews, observations, and examination of documents.

▶ 4. Focus Group Research

- ▶ Focus group research involves bringing together a small group of people to discuss a particular topic or product. Furthermore, the group is usually moderated by a researcher and the discussion is recorded for later analysis.

▶ 5. Ethnographic Research

- ▶ Ethnographic research involves conducting detailed observations of a particular culture or community. It is often used to gain a deep understanding of the beliefs, behaviors, and practices of a particular group.

Cross sectional studies and longitudinal studies.

Cross-sectional study

- ▶ Both the cross-sectional and the longitudinal studies are observational studies. This means that researchers record information about their subjects without manipulating the study environment. In our study, we would simply measure the cholesterol levels of daily walkers and non-walkers along with any other characteristics that might be of interest to us. We would not influence non-walkers to take up that activity, or advise daily walkers to modify their behaviour. In short, we'd try not to interfere.
- ▶ The defining feature of a cross-sectional study is that it can compare different population groups at a single point in time. Think of it in terms of taking a snapshot. Findings are drawn from whatever fits into the frame.
- ▶ To return to our example, we might choose to measure cholesterol levels in daily walkers across two age groups, over 40 and under 40, and compare these to cholesterol levels among non-walkers in the same age groups. We might even create subgroups for gender. However, we would not consider past or future cholesterol levels, for these would fall outside the frame. We would look only at cholesterol levels at one point in time.
- ▶ The benefit of a cross-sectional study design is that it allows researchers to compare many different variables at the same time. We could, for example, look at age, gender, income and educational level in relation to walking and cholesterol levels, with little or no additional cost.
- ▶ However, cross-sectional studies may not provide definite information about cause-and-effect relationships. This is because such studies offer a snapshot of a single moment in time; they do not consider what happens before or after the snapshot is taken. Therefore, we can't know for sure if our daily walkers had low cholesterol levels before taking up their exercise regimes, or if the behaviour of daily walking helped to reduce cholesterol levels that previously were high.

Longitudinal Study

- ▶ A longitudinal study, like a cross-sectional one, is observational. So, once again, researchers do not interfere with their subjects. However, in a longitudinal study, researchers conduct several observations of the same subjects over a period of time, sometimes lasting many years.
- ▶ The benefit of a longitudinal study is that researchers are able to detect developments or changes in the characteristics of the target population at both the group and the individual level. The key here is that longitudinal studies extend beyond a single moment in time. As a result, they can establish sequences of events.
- ▶ To return to our example, we might choose to look at the change in cholesterol levels among women over 40 who walk daily for a period of 20 years. The longitudinal study design would account for cholesterol levels at the onset of a walking regime and as the walking behaviour continued over time. Therefore, a longitudinal study is more likely to suggest cause-and-effect relationships than a cross-sectional study by virtue of its scope.
- ▶ In general, the research should drive the design. But sometimes, the progression of the research helps determine which design is most appropriate. Cross-sectional studies can be done more quickly than longitudinal studies. That's why researchers might start with a cross-sectional study to first establish whether there are links or associations between certain variables. Then they would set up a longitudinal study to study cause and effect.

Experimental Design : Definition

- ▶ In Statistics, the experimental design or the design of experiment (DOE) is defined as the design of an information-gathering experiment in which a variation is present or not, and it should be performed under the full control of the researcher. This term is generally used for controlled experiments. These experiments minimise the effects of the variable to increase the reliability of the results. In this design, the process of an experimental unit may include a group of people, plants, animals, etc.

Classification/Types of Experimental Designs

There are different types of experimental designs of research. They are:

1. Formal & Informal Research Design
2. Pre-experimental Research Design
3. True-experimental Research Design
4. Quasi-Experimental Research Design
5. Statistical experimental design.
6. Randomised Block Design

Formal Research

- ▶ Formal research is a type of research study conducted using a systematic approach and scientific methods. A formal research study typically involves several components: abstract, introduction, literature review, research design and method, results and analysis, conclusion, bibliography. All in all, formal research is more scientific in nature than informal research and aims to discover new information or solve a problem. It always uses up to date, reliable sources. Moreover, the sources of information used in research are also compiled using a reference system.
- ▶ Researchers usually begin a formal research study with a hypothesis; then, they test this hypothesis rigorously. They also explore and analyze the literature already available on their research subject. This allows them to study the research subject from multiple perspectives, acknowledging different problems that need to be solved. In comparison to informal research, formal research tends to be objective or unbiased. This mainly happens because formal research uses scientific methods which involve objective information. If the research involves participants, for example, in surveys or interviews, the researcher has to obtain their written consent first. Furthermore, we can apply the findings from formal research to a larger audience.

Informal Research

- ▶ Informal research is the use of nonscientific methods to gather and analyze data. Since the research uses nonscientific methods, its finding cannot be applied to a wider audience. Most of us conduct informal research on a regular basis. For example, conducting a simple survey, informal interviews with a target audience, seeking opinions of colleagues, searching through company files, etc. are all nonscientific methods of gathering data and are therefore forms of informal research.
- ▶ Informal research does not take much time or effort. This usually involves just looking through some information or gathering some data. For example, suppose you are going to buy a new laptop. You will ask for the opinions of your colleagues and may look through some websites and reviews to know more information about laptop brands. Your decision here will be **based** on informal research. But it's important to know this information may be subjective. Moreover, as this is an informal research, you might also not worry about the source of information.

Pre-experimental Research Design

- ▶ The simplest form of experimental research design in Statistics is the pre-experimental research design. In this method, a group or various groups are kept under observation, after some factors are recognised for the cause and effect. This method is usually conducted in order to understand whether further investigations are needed for the targeted group. That is why this process is considered to be cost-effective. This method is classified into three types, namely,
 - Static Group Comparison
 - One-group Pretest-posttest Experimental Research Design
 - One-shot Case Study Experimental Research Design

True-experimental Research Design

This is the most accurate form of experimental research design as it relies on the statistical hypothesis to prove or disprove the hypothesis. This is the most commonly used method implemented in Physical Science. True experimental research design is the only method that establishes the cause and effect relationship within the groups. The factors which need to be satisfied in this method are:

- Random variable
- Variable can be manipulated by the researcher
- Control Groups (A group of participants are familiar with the experimental group, but the experimental rules do not apply to them)
- Experimental Group (Research participants where experimental rules are applied)

Quasi-Experimental Design

- ▶ A quasi-experimental design is similar to a true experimental design, but there is a difference between the two.
- ▶ In a true experiment design, the participants of the group are randomly assigned. So, every unit has an equal chance of getting into the experimental group.
- ▶ In a quasi-experimental design, the participants of the groups are not randomly assigned. So, the researcher cannot make a cause or effect conclusion. Thus, it is not possible to assign the participants to the group.
- ▶ Apart from these types of experimental design research in statistics, there are other two methods used in the research process such as randomized block design and completely randomized design.

Statistical experimental design.

The (statistical) design of experiments (DOE) is an efficient procedure for planning experiments so that the data obtained can be analyzed to yield valid and objective conclusions. DOE begins with determining the objectives of an experiment and selecting the process factors for the study.

4 types of experimental design statistics

Four major design types with relevance to user research are experimental, quasi-experimental, correlational and single subject. These research designs proceed from a level of high validity and generalizability to ones with lower validity and generalizability. the statistical design process?

The Statistical Process has five steps:

- Design the study
- Collect the data
- Describe the data
- Make inferences
- Take action.

In a designed experiment, researchers control the conditions of the study. In an observational study, researchers don't control the conditions but only observe what happens.

Randomised Block Design

The randomised block design is preferred in the case when the researcher is clear about the distinct difference among the group of objects. In this design, the experimental units are classified into subgroups of similar categories. Those groups are randomly assigned to the group of treatment. The blocks are classified in such a way in the variability within each block should be less than the variability among the blocks. This block design is quite efficient as it reduces the variability and produces a better estimation.

Example:

In a drug testing experiment, the researcher believes that age is the most significant factor. So he divides the units according to the age groups such as

- ❖ Under 15 years old
- ❖ 15 – 35 years old
- ❖ 36 – 55 years old
- ❖ Over 55 years old

Completely Randomised Design

Of all the types, the simplest type of experimental design is the completely randomized design, in which the participants are randomly assigned to the treatment groups. The main advantage of using this method is that it avoids bias and controls the role of chance. This method provides a solid foundation for Statistical analysis as it allows the use of probability theory.

THANK YOU

Research has been called good
business, a necessity, a
gamble, a game. It is none of
these - it's a state of mind.

— Martin H. Fischer

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Module-3

SAMPLING:

Syllabus

- ▶ **Sampling:** Concepts, Types of Sampling,
Probability Sampling: simple random sampling,
systematic sampling, stratified random sampling,
cluster sampling,
- ▶ **Non Probability Sampling:** convenience
sampling- judgmental sampling, snowball
sampling, quota sampling, Errors in sampling.

Sampling: Concepts

- ▶ **What is Sampling?**
- ▶ Sampling is a process in which the fixed numbers of observations are taken randomly from a larger population. A technique which is fundamental for behavioral research is known as sampling and without using it, research work is not possible. It is impracticable and Impossible to study the whole population due to practical limitations of cost, time and other factors that are indispensable and operative in studying the whole population. For the sole reason of making the research findings economical and accurate, the concept of sampling has been introduced.
- ▶ **For example,** for taking decision about to purchase or not to purchase the fruits, a fruit merchant inspects randomly only a few of them rather than examining each and every fruit. The important objective of sampling is to obtain maximum information of the population under study using minimum of money. labour and time.

Sampling in Research:

A definite plan to obtain a sample from the sampling frame is sample design. The method or technique which is adopted by the researcher in selecting the units of sampling from the population is called sampling design.

Definition of Sampling

According to Davis S. Fox :

"In the social science, it is not possible to collect data from every respondent relevant to our study but only from some fractional part is called sampling".

According to Cochran :

"In every branch of science we lack the resources, to study more than a fragment of the phenomena that might advance our knowledge". In this definition a fragment is the sample and phenomena is the population.

Basic Concepts of Sampling

1) Universe/Population :

i) Finite Universe :

ii) Infinite Universe :

2) Statistical Population :

3) Sample :

4) Census :

5) Sampling Frame :

6) Sample Unit :

7) Sample Design :

Characteristics of a Good Sample frame

7

A good sampling frame should be :

- **Relevant** : Research topic must be directly linked with relevant things.
- **Complete** : Coverage of all relevant items.
- **Precise** : It should exclude all the irrelevant items.
- **Up-to-Date** : Incorporating recent changes and additions, and deleting redundant items.

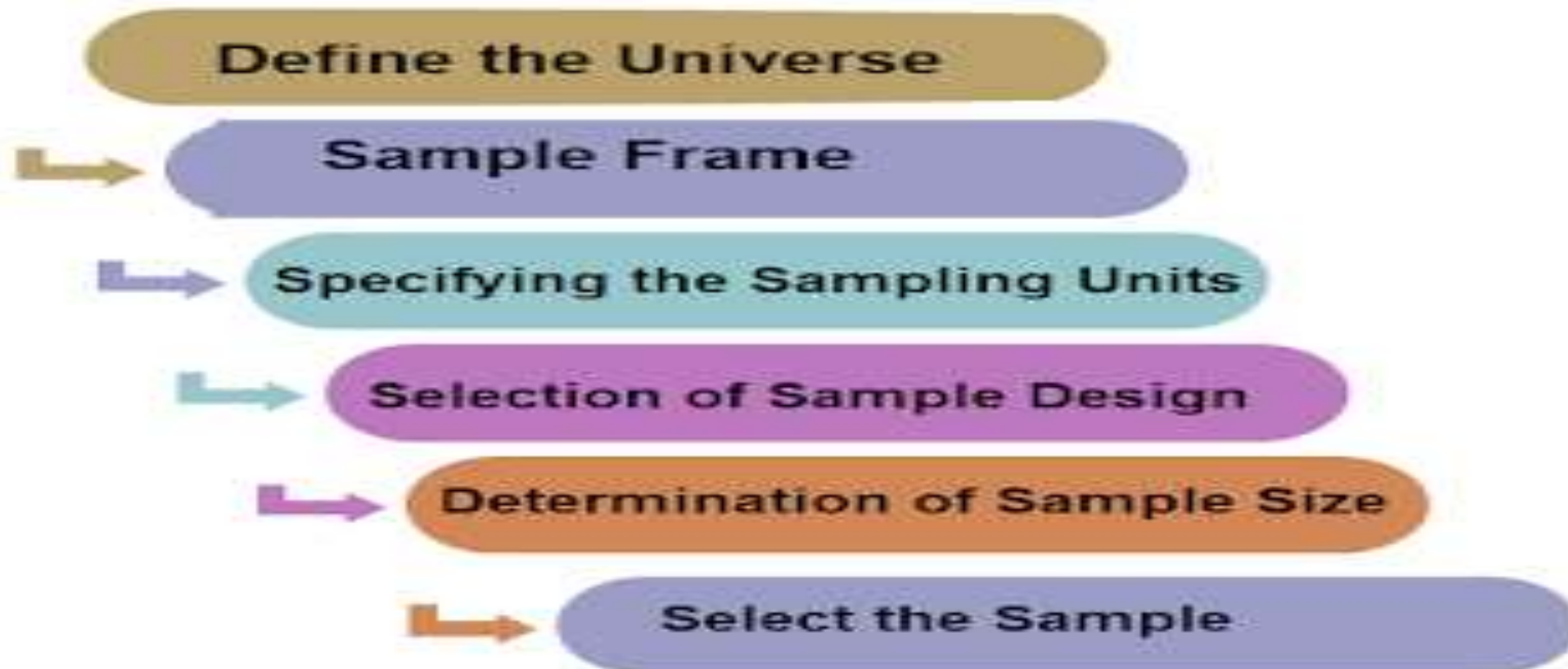
Characteristics of a Good Sample

The various features of a good sample design are as follows :



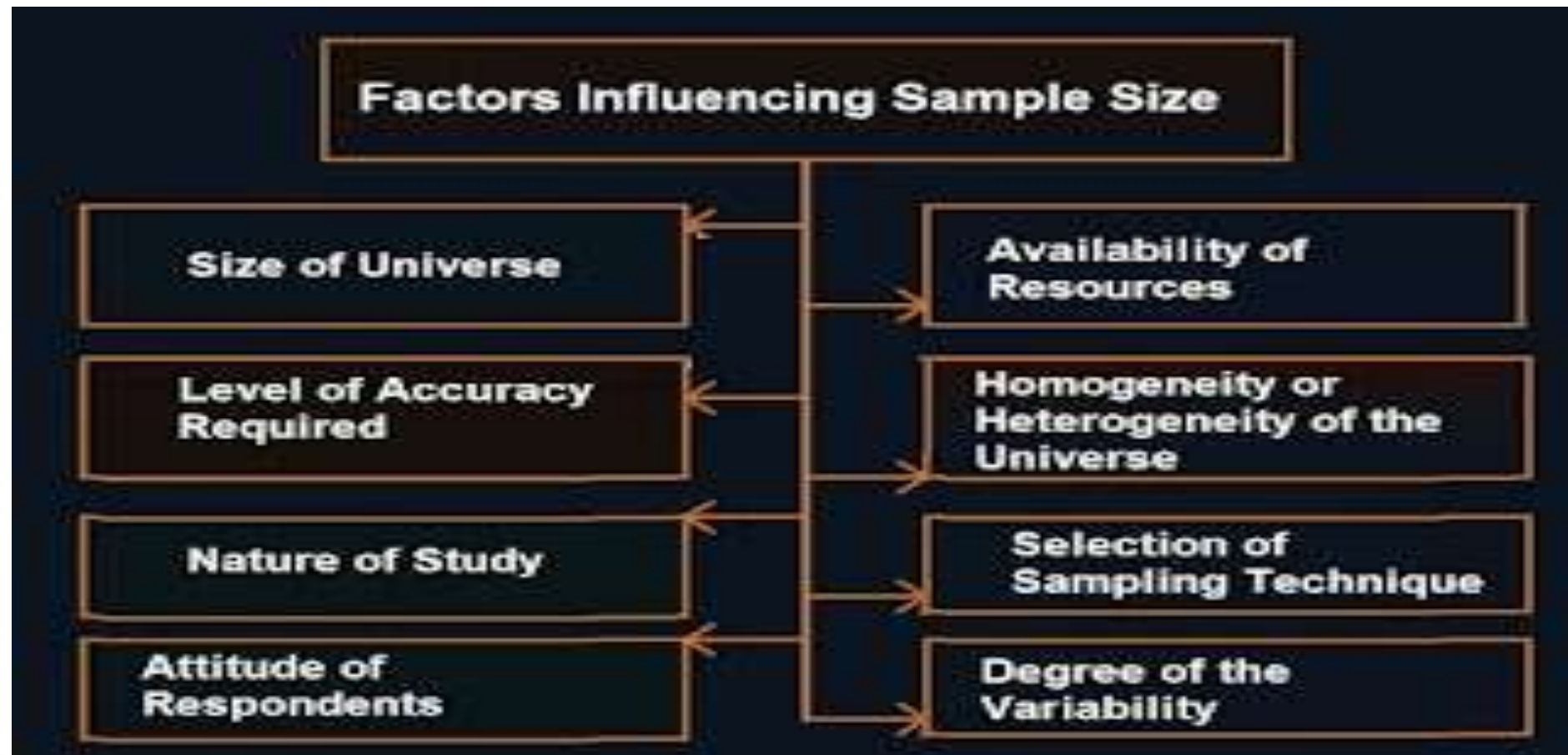
Procedure of Sampling

Figure : Sampling Procedure



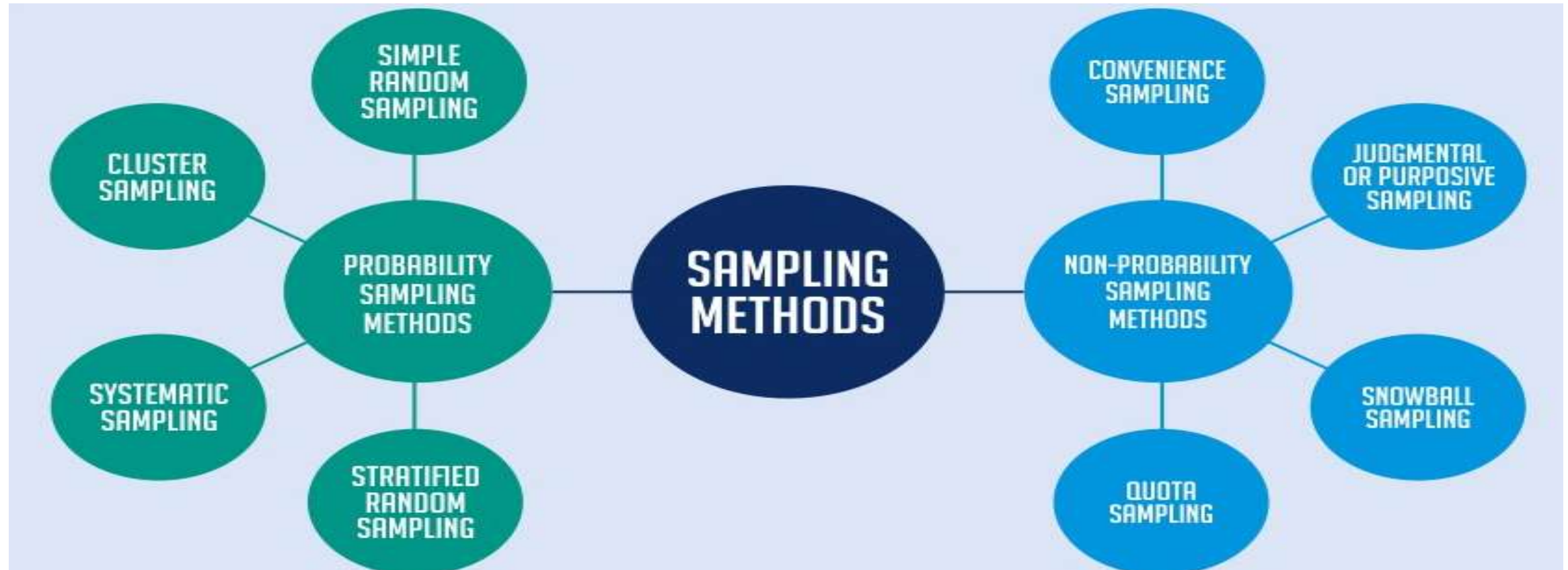
Factors Influencing Sample Size/ Sample Size Constraints

10



Types/ Methods of Sampling

11



Probability Sampling:

Probability sampling refers to the selection of a sample from a population, when this selection is based on the principle of randomization, that is, random selection or chance. Probability sampling is more complex, more time-consuming and usually more costly than non-probability sampling.

Simple random sampling

Simple random sampling is a statistical method in which everyone in a population has an equal chance of being selected into a sample. The sample represents a smaller and more manageable portion of the people that can be studied and analyzed. It's a fundamental technique to gather data and make inferences about a population.

Two approaches aim to minimize any biases in the process of this method:

Method of lottery

Using the lottery method is one of the oldest ways and is a mechanical example of random sample. Researchers draw numbers from the box randomly to choose samples. In this method, the researcher gives each member of the population a number.

Use of random numbers

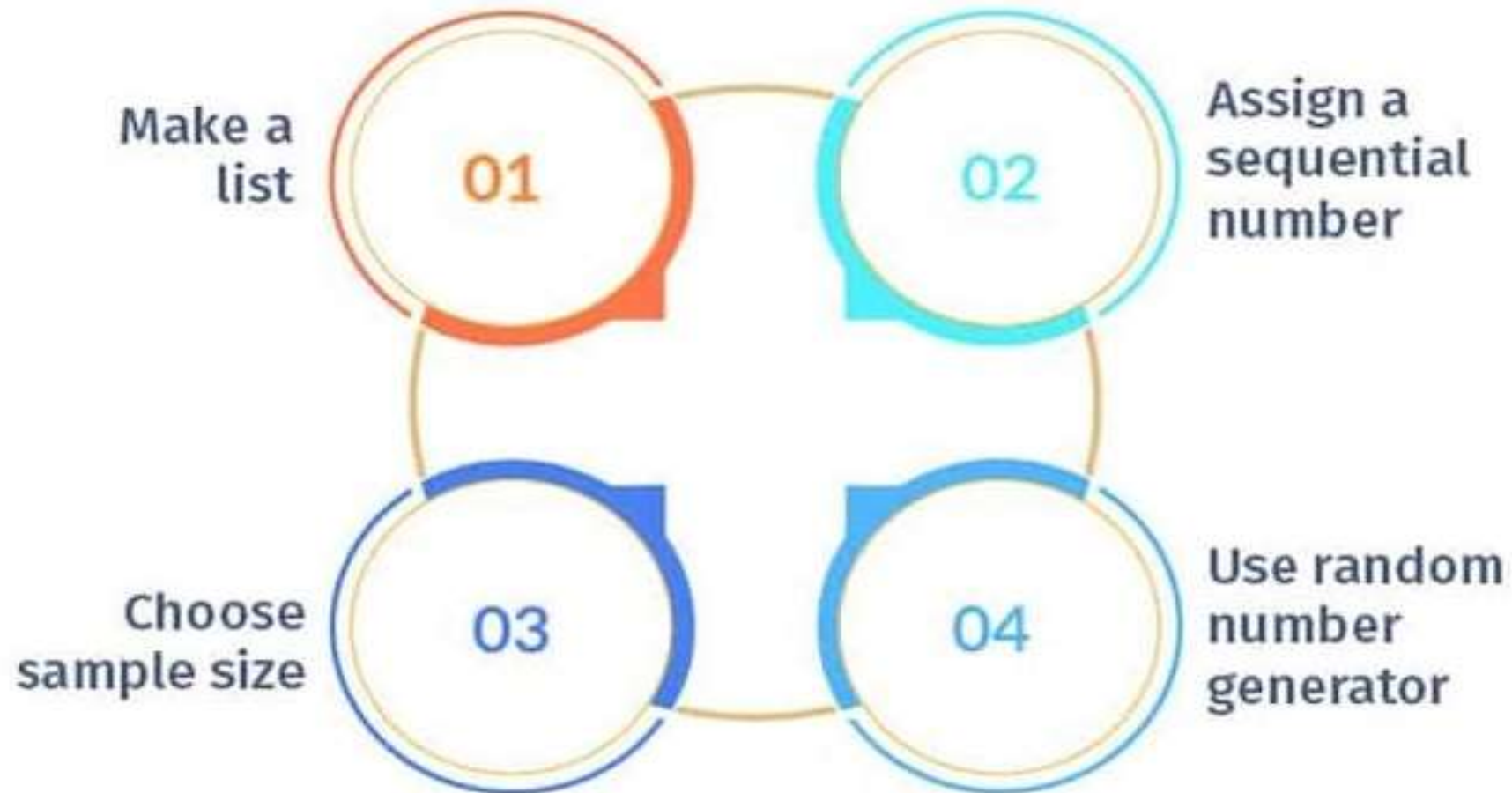
Using random numbers is an alternative method that also involves numbering the population. A numbered table similar to the one below can help with this sampling technique.

Using random numbers is an alternative method that also involves numbering the population. A numbered table similar to the one below can help with this sampling technique.

Random Number Table

1	69	24	40	68	29	39	95	60	30
97	23	70	59	79	4	47	19	38	20
13	44	5	71	12	99	78	34	9	96
34	55	83	21	72	3	37	85	61	2
22	80	18	82	54	32	84	16	46	88
7	43	6	48	11	92	63	53	86	28
56	90	36	91	64	45	15	73	10	87
49	65	50	14	51	33	89	52	74	57
98	17	100	58	5	8	77	25	62	31
27	76	66	81	26	93	41	94	67	42

STEPS TO CONDUCT SIMPLE RANDOM SAMPLING



Systematic sampling definition

Systematic sampling is a probability sampling method where the researcher chooses elements from a target population by selecting a random starting point and selecting sample members after a fixed 'sampling interval.'

What are the steps to form a sample using the systematic sampling technique?

Here are the steps to form a systematic sample:

Step one: Develop a defined structural audience to start working on the sampling aspect.

Step two: As a researcher, figure out the ideal size of the sample, i.e., how many people from the entire population to choose to be a part of the sample.

Step three: Once you decide the sample size, assign a number to every member of the sample.

Step four: Define the interval of this sample. This will be the standard distance between the elements.

For example, the sample interval should be 10, which is the result of the division of 5000 (N =size of the population) and 500 (n =size of the sample).

Step five: Select the members who fit the criteria which in this case will be 1 in 10 individuals.

Step six: Randomly choose the starting member (r) of the sample and add the interval to the random number to keep adding members in the sample. r , $r+i$, $r+2i$, etc. will be the elements of the sample.

Systematic Sampling Types

- Systematic random sampling
- Linear systematic sampling
- Circular systematic sampling

Stratified random sampling

Stratified random sampling is a widely used statistical technique in which a population is divided into different subgroups, or strata, based on some shared characteristics. The purpose of stratification is to ensure that each stratum in the sample and to make inferences about specific population subgroups.

This technique is beneficial in cases where the population is heterogeneous, and a simple random sample may not provide accurate results. By dividing the population into strata, researchers can ensure that their sample is representative of the population and avoid sampling biases.

8 Steps to Conduct Stratified Random Sampling

20



Types of Stratified Random Sampling

- Proportionate Sampling:
- Disproportionate Sampling:

What is Cluster Sampling?

Cluster sampling is a probability sampling technique where researchers divide the population into multiple groups (clusters) for research. So researchers then select random groups with a simple random or systematic random sampling technique for data collection and unit of analysis.

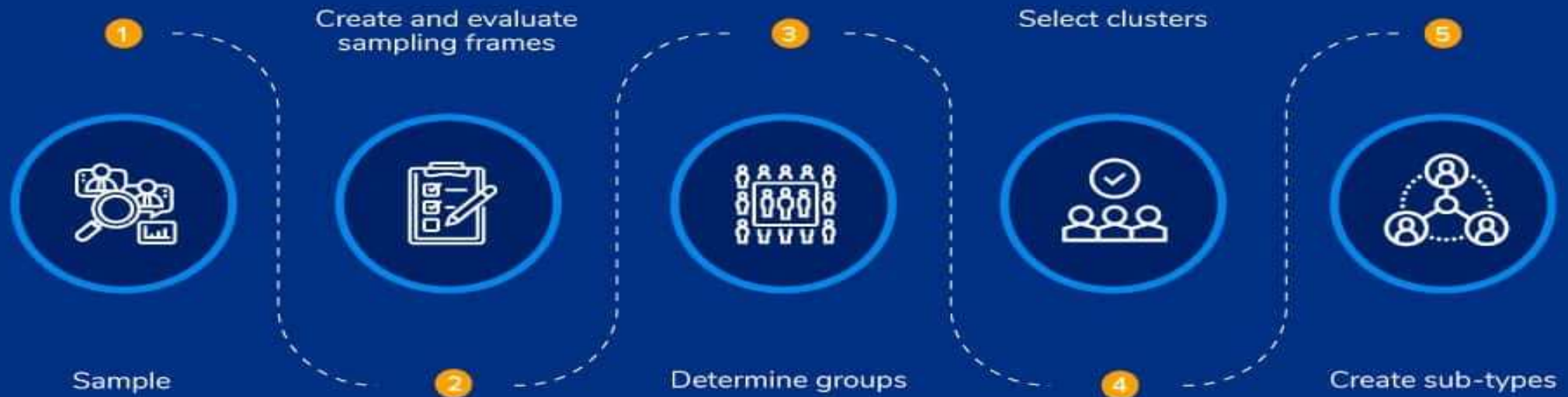
Example: A researcher wants to conduct a study to judge the performance of sophomores in business education across the U.S. It is impossible to conduct a research study that involves a student in every university. Instead, cluster sampling allows the researcher to club the universities from each city into one cluster. These clusters then define the sophomore student population in the U.S. Next, either using simple random sampling or systematic random sampling and randomly picking clusters for the research study. Subsequently, by using simple or systematic sampling, the sophomore's from each of these selected clusters can be chosen on whom to conduct the research study.

Types of Cluster Sampling

- Single-stage Cluster Sampling:
- Two-stage Cluster Sampling:
- Multiple-stage Cluster Sampling:

Steps to conduct Cluster Sampling

Steps to conduct cluster sampling



Cluster Sampling Advantages

1



Consumes less time and cost

2



Convenient access

3



Data accuracy

4



Ease of implementation

Non Probability Sampling:

Non-probability sampling is a method of selecting units from a population using a subjective (i.e. non-random) method. Since non-probability sampling does not require a complete survey frame, it is a fast, easy and inexpensive way of obtaining data.

What is convenience sampling?

Convenience sampling is defined as a method adopted by researchers where they collect market research data from a conveniently available pool of respondents.

It is the most commonly used sampling as it's incredibly prompt, uncomplicated, and economical. Members are often readily approachable to be a part of the sample.

IMPORTANCE OF CONVENIENCE SAMPLING

Firstly

Cost-effective and time-efficient method of data collection



Second

Beneficial for companies seeking input from their customers or stakeholders

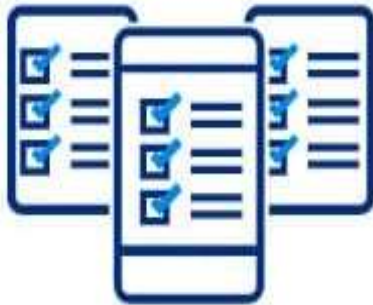
Advantages of using convenience sampling

29



Hacks to analyze convenience sampling data

1



Take multiple samples. It helps you in producing reliable results.

2



Repeat the survey to understand whether your results truly represent the population.

3



Try cross-validation for half the data. Then compare the findings with the other half of the data.

Judgmental sampling

Judgmental sampling, also called purposive sampling or authoritative sampling, is a non-probability sampling technique in which the sample members are chosen only on the basis of the researcher's knowledge and judgment. As the researcher's knowledge is instrumental in creating a sample in this sampling technique, there are chances that the results obtained will be highly accurate with a minimum margin of error.

Consumes minimum time
for execution

Directly approachable
respondents



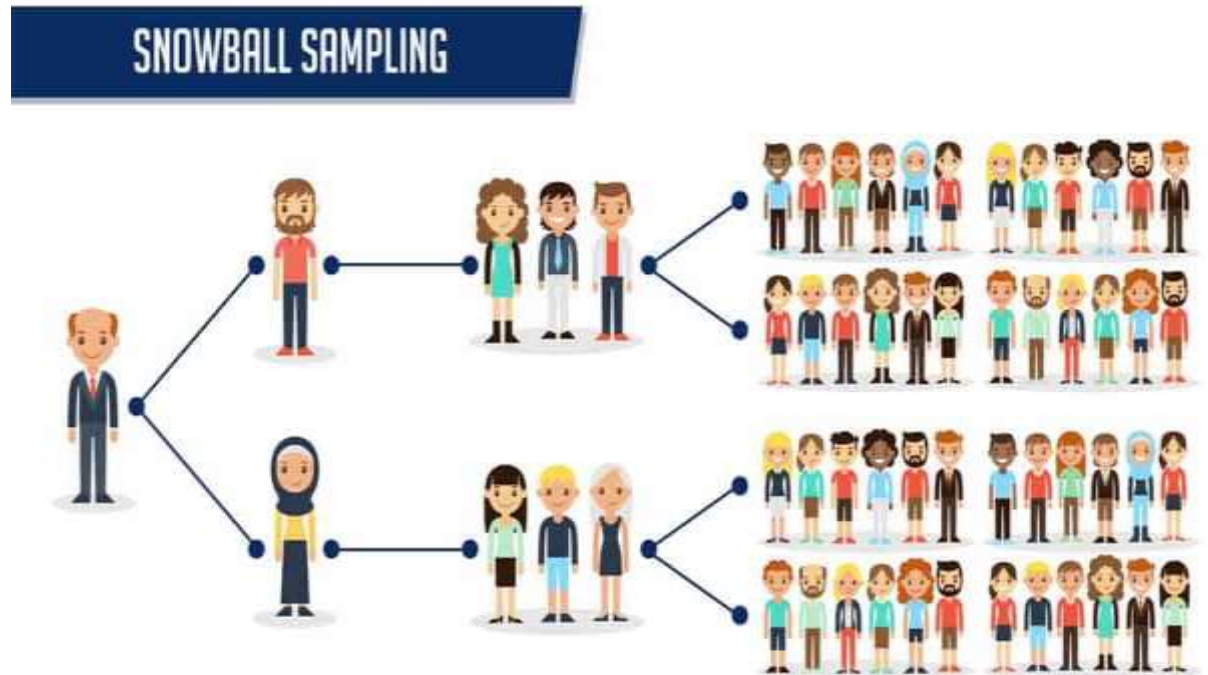
Judgmental
Sampling

Almost real-time
results

Easy
to conduct

Snowball Sampling:

Snowball sampling or chain-referral sampling is defined as a non-probability sampling technique in which the samples have rare traits. This is a sampling technique, in which existing subjects provide referrals to recruit samples required for a research study.



Types of Snowball Sampling

1. **Linear Snowball Sampling:** The formation of a sample group starts with one individual subject providing information about just one other subject and then the chain continues with only one referral from one subject. This pattern is continued until enough number of subjects are available for the sample.
2. **Exponential Non-Discriminative Snowball Sampling:** In this type, the first subject is recruited and then he/she provides multiple referrals. Each new referral then provides with more data for referral and so on, until there is enough number of subjects for the sample.
3. **Exponential Discriminative Snowball Sampling:** In this technique, each subject gives multiple referrals, however, only one subject is recruited from each referral. The choice of a new subject depends on the nature of the research study.

Quota sampling:

Quota sampling is defined as a non-probability sampling method in which researchers create a convenience sample involving individuals that represent a population. Researchers choose these individuals according to specific traits or qualities. They decide and create quotas so that the market research samples can be useful in collecting data. These samples can be generalized to the entire population. The final subset will be decided only according to the interviewer's or researcher's knowledge of the population.

For example, a cigarette company wants to find out what age group prefers what brand of cigarettes in a particular city. They apply survey quota on the age groups of 21-30, 31-40, 41-50, and 51+. From this information, the researcher gauges the smoking trend among the population of the city.

Types of quota sampling:

- Controlled quota sampling:
- Uncontrolled quota sampling:

Characteristics of quota sampling



Aims to get the best representation of respondents in the final sample.

Saves research costs if the quotas accurately represent the population.



Quotas replicate the population of interest in a real sense.

It monitors the subgroups of individuals who take the survey.



The estimates produced are more representative.

The researcher always divides the population into subgroups.



The quality of quota samples vary.

The sample represents the entire population.



Saves research data collection time as the sample represents the population.

Researchers use the sampling method to identify the traits of a specific group of people.



Errors in sampling.

Assume you are a market researcher of a company looking to introduce a new product to the market. You must collect data from a sample of potential customers as part of your research to determine their preferences and purchasing behavior. But how can you be sure that the information you get from your sample is accurate for all of the people who might buy your product? The idea of sampling error comes into play here.

It is the difference between what a sample has and what the entire population has. It can significantly affect how accurate and reliable market research data is.

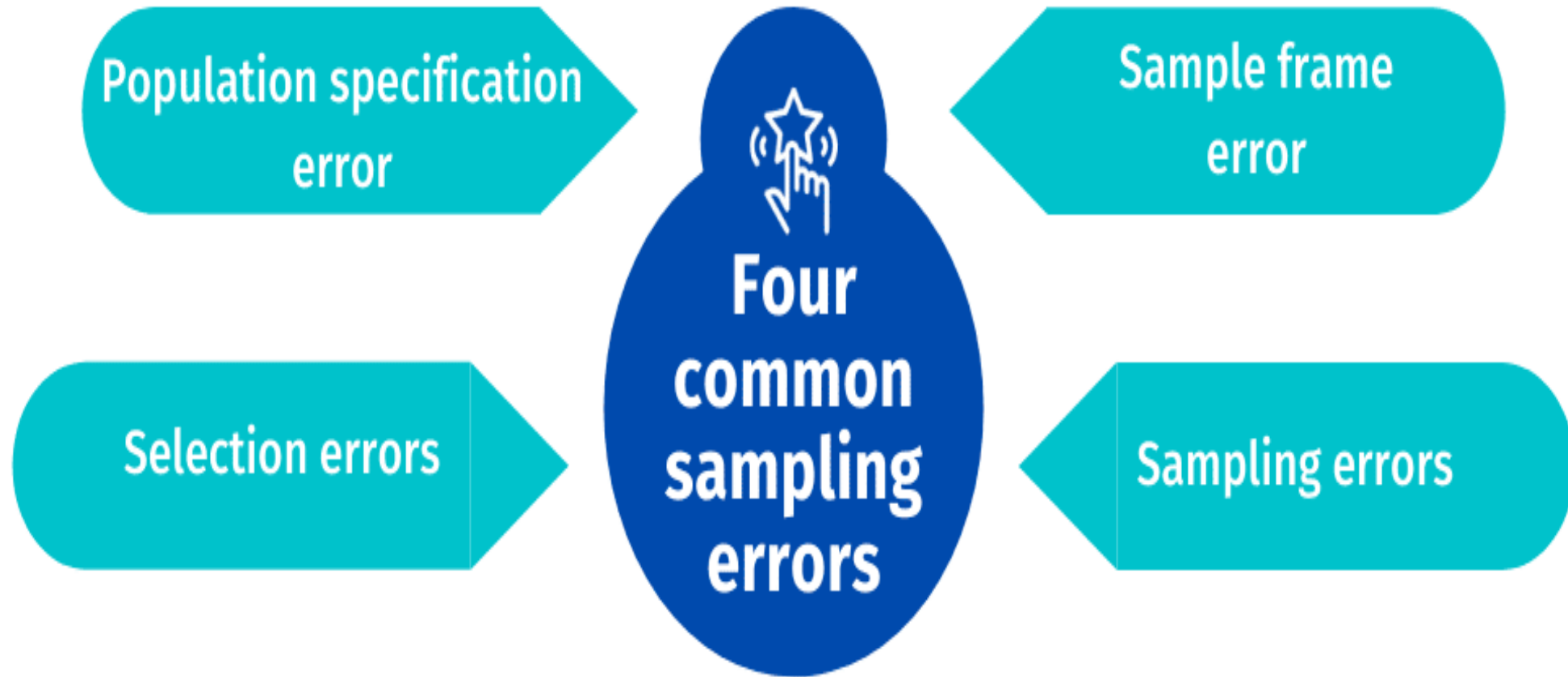
We will cover ways to reduce sampling error in this article to get more accurate and reliable results. Now, grab your favorite coffee and get ready to explore what sampling error is.

What is a sampling error?

A sampling error occurs when the sample used in the study does not represent the entire population. Although sampling errors occur frequently, researchers always include a margin of error in their conclusions as a matter of statistical practice.

The margin of error is the amount allowed for a miscalculation to represent the difference between the sample and the actual population.

Sampling is a type of analysis where a small sample of observations is chosen from a larger population. The selection process can produce both sampling errors and non-sampling errors.



Steps to Reduce Sampling Errors

1



Increase
sample size

2



Divide the population
into groups

3



Know
your population

THANK YOU



Sampling, statisticians have told us,
is a much more effective way of
getting a good census.

— Rob Lowe —



Module-4

Data Collection:



Syllabus

Data Collection: Meaning, types, **Data collection methods:** Observations, survey and interview

techniques, **Questionnaire design:** Meaning, process of designing questionnaire. Qualitative Techniques of data collection Secondary data Sources: advantages and disadvantages.

Measurement and Scaling Techniques: Basic measurement scales-Nominal scale, Ordinal scale, Interval scale, Ratio scale. Attitude measurement scale - Likert Scale, Semantic Differential Scale, Thurston scale,

Multi-Dimensional Scaling: Non comparative scaling techniques.

Data Collection: Meaning

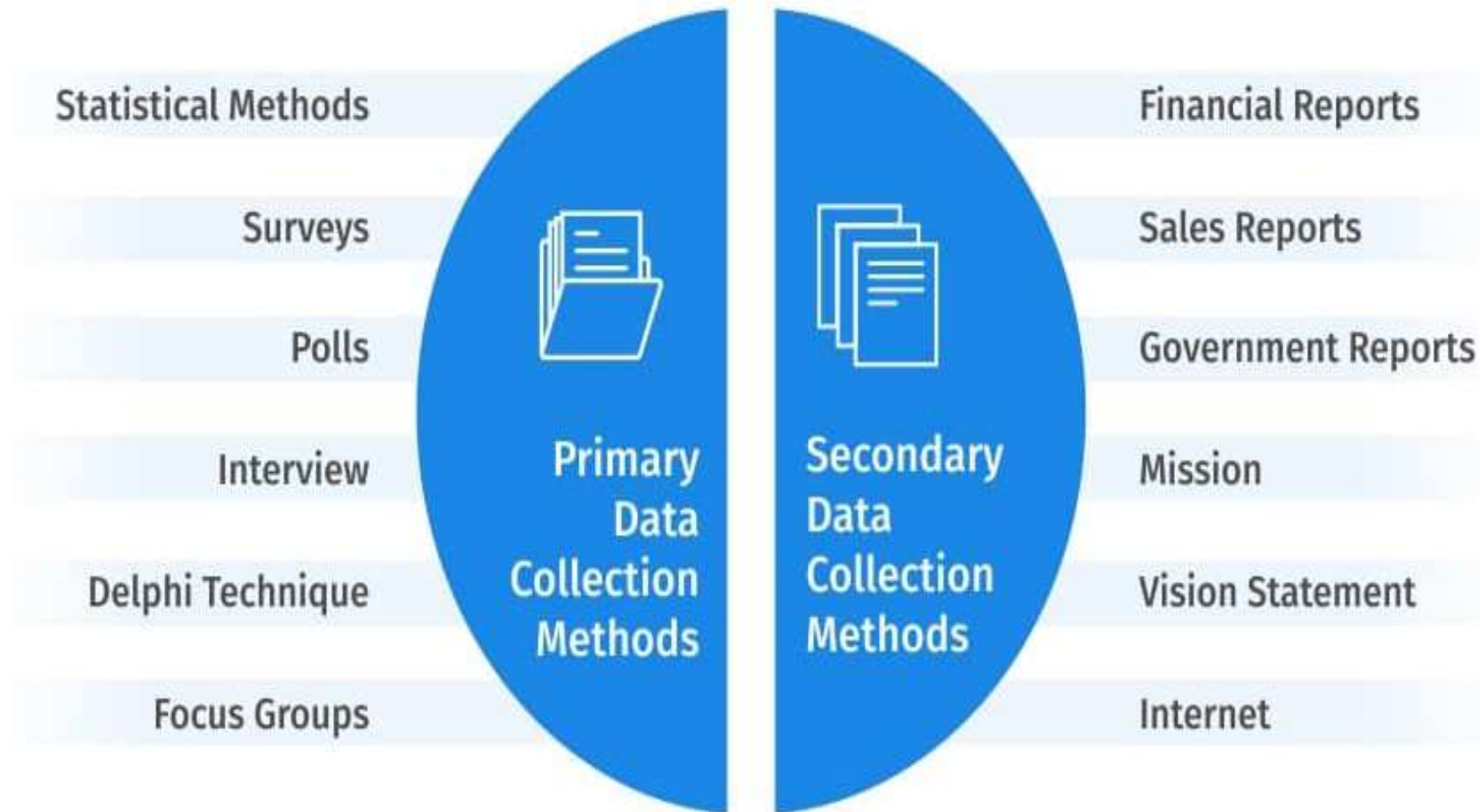
Data is a collection of facts, figures, objects, symbols, and events gathered from different sources. **Organizations collect data with various data collection methods to make better decisions.** Without data, it would be difficult for organizations to make appropriate decisions, so data is collected from different audiences at various points in time.

Data collection methods are techniques and procedures used to gather information for research purposes. These methods can range from simple self-reported surveys to more complex experiments and can involve either quantitative or qualitative approaches to data gathering.

Importance of Data Collection Methods

- Determines the quality and accuracy of collected data.
- Ensures that the data is relevant, valid, and reliable.
- Helps reduce bias and increase the representativeness of the sample.
- Essential for making informed decisions and accurate conclusions.
- Facilitates achievement of research objectives by providing accurate data.
- Supports the validity and reliability of research findings.

Types of Data Collection Methods



Quantitative Methods:

Quantitative techniques for market research and demand forecasting usually use statistical tools. In these techniques, demand is forecasted based on historical data. These methods of primary data collection are generally used to make long-term forecasts. Statistical methods are highly reliable as subjectivity is minimal in these methods.



Qualitative Methods:

Qualitative data collection methods are especially useful in situations when historical data is not available. Or there is no need of numbers or mathematical calculations. Qualitative research is closely associated with words, sounds, feeling, emotions, colors, and other elements that are non-quantifiable. These techniques are based on experience, judgment, intuition, conjecture, emotion, etc.

Qualitative Methods



Surveys



Polls



Interviews



Delphi Technique



Focus Groups



Questionnaire

Observation Method

- **Meaning and examples**
- **Observational research** is used in cases where you want to avoid an error that can be a result of bias during evaluation and interpretation processes. Observational research is one of the many other research methods to obtain objective data by watching a participant and recording it for analysis at a later stage.
- A researcher can use observational research, for example in a school, and record the behavior of the children at a young age. Are the children comfortable sharing their tiffin at such an early age will make a good study for the researcher? In this example, the researcher can observe and record the details objectively. The observational research data collection method is associated with observational research methods but with a few ethical issues as it needs the full consent of a research participant.

The observation method in data collection can be

- **Structured observation method** – Structured Observation is a systematic observation method where data is collected as per a pre-defined schedule. The specific variable is used in this observation data collection method.
- **Unstructured observation method** – The unstructured observation method is different from structured observation because it is conducted in a free and open manner without using any pre-determined objectives, schedules, or variables.

Sampling Methods for Observational Data

- **Event sampling** – In the event sampling observation method the researcher decides beforehand what events or behavior he will record and which ones he is going to ignore
- **Time sampling** – In the time sampling observation method, the researcher chooses the time when he will observe. He makes a record of the occurrence only in the specified and pre-determined period
- **Target-time or instantaneous sampling** – In the target-time or instantaneous sampling observation method, the researcher decides beforehand the moments when the observation will happen and will be recorded at that moment. Everything that happens before or after that moment is of no consequence and hence is typically ignored

Types of Observational Research Methods

- 1) Controlled observations
- 2) Naturalistic observations
- 3) Participant observation
 - *Overt participant observation:*
 - *Covert participant observation:*

Advantages of the Observation Method

- Provides direct access to research phenomena
- By observing firsthand, the researcher can collect, check and record accurate data
- Greater flexibility in terms of application
- Generate a permanent record of phenomena and the researcher or others can refer with it later
- The organization method is one of the simplest methods of data collection. It does not require too much technical knowledge
- The observation method is one of the best ways to formulate a hypothesis. The researcher can observe and come to know about the activities, perceptions, likes, and dislikes to form a theory on his subject
- Observational methods are one of the most common methods used in all sciences and is very easy to follow and accept
- In some instances observation is the only available tool to collect essential data and information
- The observation method does not require the willingness of the participant to record. The researcher can observe from a distance and record his findings

Disadvantages of the Observation Method

- Observational methods Face a severe disadvantage because it takes a longer time frame compared to other data collection methods
- There is a chance of higher observer bias in the observation method
- Several personal behaviors are not open for observation and this proves a limitation in case of observation method
- There is a higher chance of the observer influencing the behavior of a sample group elements
- Uncertainties of the event cannot determine the actual time when the event will take place, and this is why every occurrence that is open to observation cannot be observed
- Many of the incidents are abstract like love, affection and the researcher can't gain an exact and correct account of those
- The social phenomena generalization made by observation are not considered reliable as it cannot be used for lab experiments
- In some cases, it is seen that two persons observing the same phenomena come at different results and this can lead to faulty perceptions
- Observation method is considered an expensive affair as it requires hard effort, plenty of time and high cost

Survey Data Collection Methods with Examples

- The methods used to collect survey data have evolved with the change in technology. From face-to-face surveys, telephonic surveys to now online and [email surveys](#), the world of survey [data collection](#) has changed with time. Each survey data collection method has its pros and cons, and every researcher has a preference for gathering accurate information from the target sample.
- The [survey response rates](#) for each of these data collection methods will differ as their reach and impact are always different. Different ways are chosen according to specific target population characteristics and the intent to examine human nature under various situations.

Types of survey data based on deployment methods:

- **Online Surveys**
- **Face-to-face Surveys**
- **Telephone Surveys**
- **Paper Surveys**

Interview techniques

- An interview is a way to get information from a person by asking questions and hearing their answers.
- An interview is a question-and-answer session where one person asks questions, and the other person answers those questions. It can be a one-on-one, two-way conversation, or there can be more than one interviewer and more than one participant.
- The interview is the most important part of the whole selection process. It is used to decide if a person should be interviewed further, hired, or taken out of consideration. It is the main way to learn more about applicants and the basis for judging their job-related knowledge, [research skills](#), and abilities.

Fundamental Types of Interviews in Research

Fundamental Types of Interviews in Research

1



**Structured
Interviews**

2



**Semi-Structured
Interviews**

3



**Unstructured
Interviews**

OTHER TYPES OF INTERVIEWS



Behavioral



Panel



Group



Case



Technical



Stress

Methods of Research Interviews:



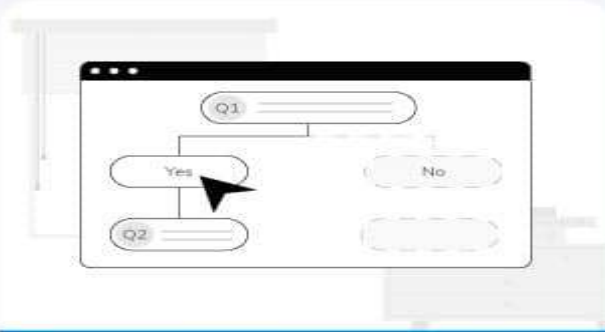
Questionnaire design:

- A questionnaire is a research instrument that consists of a set of questions or other types of prompts that aims to collect information from a respondent. A research questionnaire is typically a mix of close-ended questions and open-ended questions.
- Open-ended, long-form questions offer the respondent the ability to elaborate on their thoughts. Research questionnaires were developed in 1838 by the Statistical Society of London.
- The data collected from a data collection questionnaire can be both qualitative as well as quantitative in nature. A questionnaire may or may not be delivered in the form of a survey, but a survey always consists of a questionnaire.

Advantages of a well-designed online questionnaire



Collect a large amount of data in less time.



There is less chance of bias with the help of logic questions.



Online survey software is fast and cost effective to design, distribute and analyze.



It can be customized to reflect your brand voice.



Responses can be compared with historical data.



Respondents can answer the questionnaire without revealing their identity.

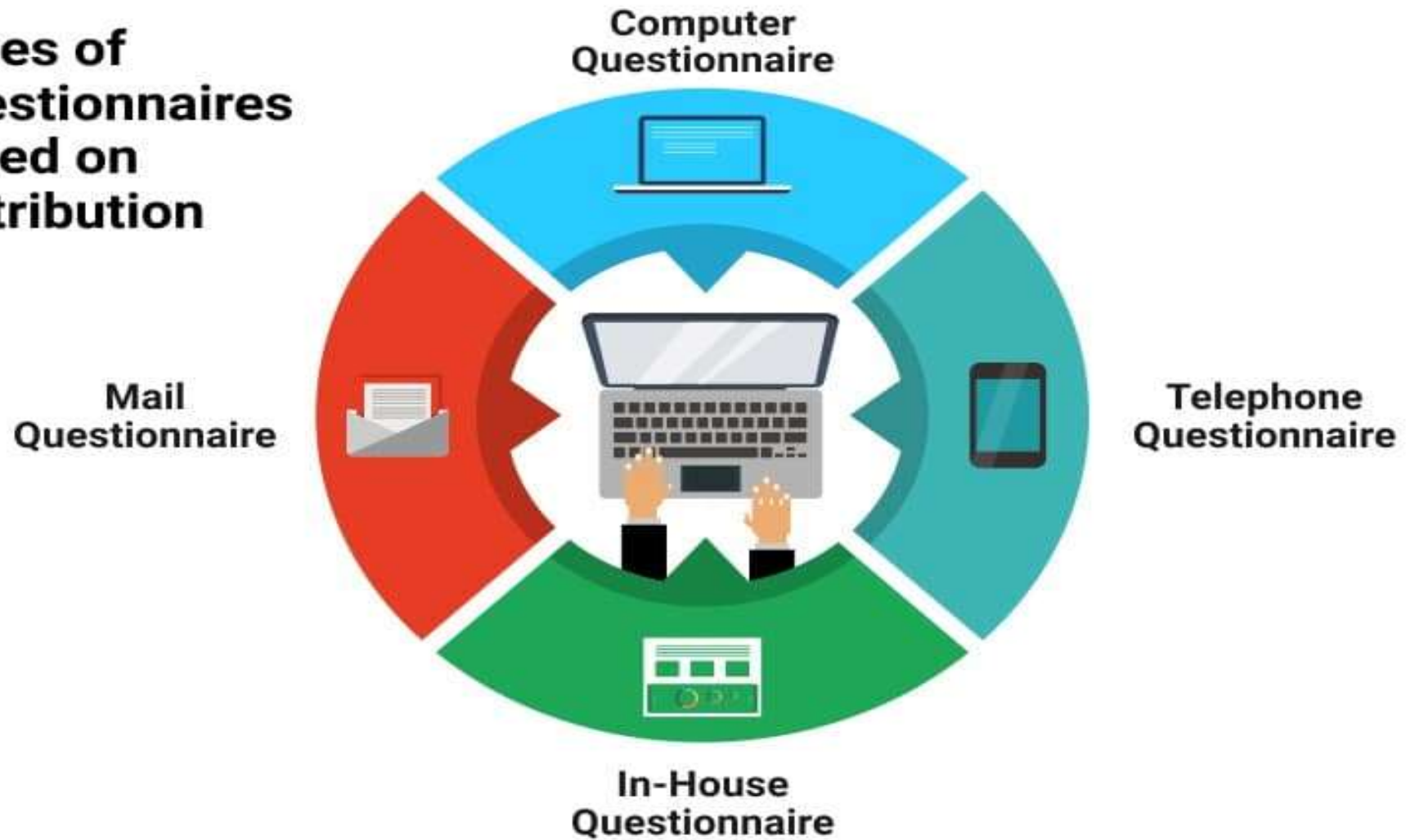
Characteristics of a good questionnaire

- Uniformity:
- Exploratory:
- Question Sequence:

Types of questions in a questionnaire

- **Structured Questionnaires:**
- **Unstructured Questionnaires:**
- **Open-Ended Questions:**
- **Dichotomous Questions: Yes/ No**
- **Multiple-Choice Questions**
- **Scaling Questions:**
- **Pictorial Questions : Multiple choice with picture**

Types of Questionnaires based on Distribution



How to design a Questionnaire



RULES OF HOW TO DESIGN A QUESTIONNAIRE



Identify what you want to **cover** in a survey



Don't **mince** your words



Ask only **one** question at a time



Be **flexible** with your options



Open/closed-ended question, it's a **tough choice**



It is important to know your **audience**



Choosing the **right tool** is important



Steps Involved in Questionnaire Design

1. Identify the scope of your research:
2. Keep it simple:
3. Ask only one question at a time:
4. Be flexible with your options:
5. The open-ended or closed-ended question is a tough choice:
6. It is essential to know your audience:
7. Choosing the right tool is essential:

Measurement and Scaling Techniques

Types of Measurement Scales

Nominal scale

It's used to label variables in different classifications and does not imply a quantitative value or order.



Ordinal Scale

It's used to represent non-mathematical ideas such as frequency, satisfaction, happiness, a degree of pain, etc.



Interval Scale

It's defined as a numerical scale where the order of the variables as well as the difference between these variables is known.



Ratio Scale

It's a variable measurement scale that not only produces the order of the variables, but also makes the difference between the known variables along with information about the value of the true zero.

Attitude measurement scale - Likert Scale

Likert scale survey question and the strongly agree to strongly disagree framework is a staple in the pantry of questions we regularly use on surveys. In the years I have been involved in constructing surveys and teaching market research I have seen many techniques and question types rise and fall in favor, yet the Likert scale holds its own.

[illegible]

Semantic Differential Scale

The ease-of-understanding and the popularity it comes with it makes it extremely reliable. The data collection is accurate due to the versatility that these [survey questions](#) come with.

Researchers use the semantic differential scale questions to ask respondents to rate products, organization, or services with multi-point questions with polar adjectives at the extremes of this scale like likely/ unlikely, happy/sad, loved the service/ hated the service.

Semantic differential scale examples & question types

1. Slider rating scale: Questions that feature a graphical slider give the respondent a more interactive way to answer the semantic differential scale question.

Which of the following categories best describes your last experience purchasing a product or service on our website?

Very Pleasant Somewhat Pleasant Neither Pleasant nor Unpleasant Somewhat Unpleasant Very Unpleasant

My experience was Somewhat Pleasant



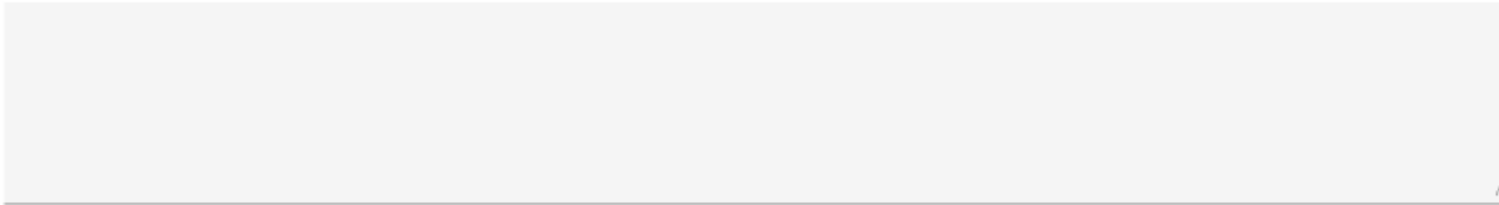
A horizontal slider bar is shown with a blue dot indicating the selected rating. The dot is positioned between 'Somewhat Pleasant' and 'Neither Pleasant nor Unpleasant', closer to 'Somewhat Pleasant'. The bar is a thick blue line that extends from the 'Somewhat Pleasant' label to the right edge of the scale.

- **2. Non-slider rating scale:** The non-slider question uses typical radio buttons for a more traditional survey look and feel. Respondents are more used to answering.

[illegible]

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- **3. Open-ended questions:** These questions give the users ample freedom to express their emotions about your organization, products, or services.

Comments/Suggestions:

A large, empty, light gray rectangular box for user input, representing an open-ended question form. It is positioned below the 'Comments/Suggestions:' label and has a thin horizontal line at its base.

-
- **4. Ordering:** The ordering questions offer the scope to rate the parameters that the respondents feel are best or worst according to their personal experiences.

Based on what you have seen, heard, and experienced, please rank the following brands according to their reliability. Place a "1" next to the brand that is most reliable, a "2" next to the brand that is next most reliable, and so on. Remember, no two cars can have the same ranking.

Ford	-- Select --	▼
BMW	-- Select --	▼
Chrysler	-- Select --	▼
Chevrolet	-- Select --	▼

5. Satisfaction rating: The easiest and eye-catching semantic differential scale questions are the satisfaction rating questions.

How satisfied are you with our services



Very Unsatisfied



Unsatisfied



Neutral



Satisfied



Very Satisfied

Thurstone Scale: Definition

Thurstone scale is defined as a unidimensional scale that is used to track respondent's behavior, attitude or feeling towards a subject.

This scale consists of statements about a particular issue or topic where each statement has a numerical value that indicates the respondents attitude towards the topic as favorable or unfavorable. Respondents indicate the statements that they agree with, and an average is computed. A mean score of the agreements or disagreements is calculated as the attitude of the respondent towards the topic.

Derive the Final Question

- Step 1 – Develop statements:
- Step 2 – Rank each statement with a panel of judges:
- Step 3 – Calculate median and/or mean and Interquartile range (IQR):
- Step 4 – Sorting the table:
- Step 5 – Selecting final variables or options:

Administering and Analysis of the Thurstone Scale Question

- **Administering and analysis by median or mean**

The question and the subsequent options can be administered to the respondents by using either the mean or the median score in the below format. The weights of the statements are summed and divided by the number of statements checked. If a respondent were to agree with statements 2, 5, 7 and 10; the attitude score is $10.5 + 2.5 + 4.5 + 6.0 = 23.5/4 = 5.8$. Dividing the number of statements puts this score at a little above the midway point of the 1-11 scale. This score indicates that the attitude is slightly favorable to diversity hiring in the organization.

- **Administering and analysis by simple count or percentage:**

In the same example above, if the question is administered without the mean or median score, the calculation can be depicted by a simple count of agreement on the 1-11 scale or a percentage. If the respondent were to agree with statements 1, 4, 5, 6, 8, 9 and 11; the agreement count is 7 out of 11 which puts the percentage at 63.63% which means that the attitude is favorable towards diversity hiring.

Characteristics of a Thurstone Scale Question

- **They are two stepped:** A Thurstone scale question is never administered at the first iteration stage without the rating by judges. This is an important characteristic of this question as the options that a respondent sees are weighted and there is consensus of them being in the [survey](#).
- **Mean or median is always calculated:** Since each option is weighted, the mean or median is calculated for each option. This also forms the basis of the selection method for use in the final survey.
- **Only agree or disagree options:** The respondent makes a selection only on the basis of the agreement or disagreement with the statement.

Use of a Thurstone Scale Survey

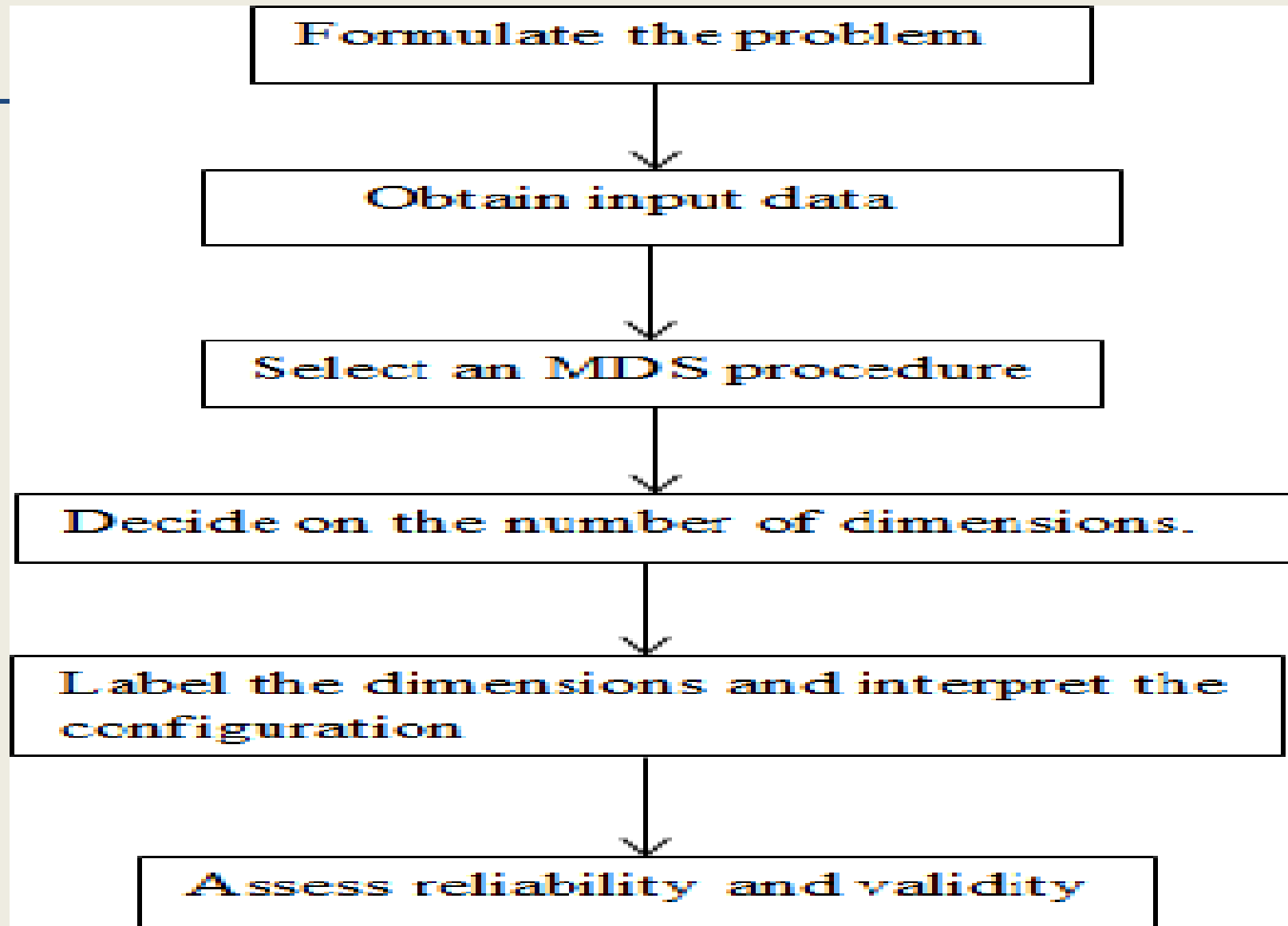
- The Thurstone scale survey is used to measure the respondents attitudes on a particular subject. The scale can be applied to a wide range of [market research surveys](#), in market research including:
- **Surveys that measure opinions:** The Thurstone scale question produces quantifiable data about the measures of strength of the respondents opinions.
- **Those that gauge attitudes or feelings:** This scale is used effectively in [customer satisfaction](#) to predict future purchasing trends and in [employee engagement](#) to calculate turnover.

Multi-Dimensional Scaling: Non comparative scaling techniques

- Multidimensional Scaling (MDS) is a class of procedures for representing perception and preferences of respondents spatially by means of a visual display. Perceived or psychological relationships among points in a multidimensional space. These geometric representations are often called spatial maps. The axes of the spatial map are assumed to denote the psychological bases or underlying dimensions respondents use to form perceptions and preferences for stimuli. MDS has been used in marketing to identify:
 - 1. The number and nature of dimensions consumers use to perceive different brands in the marketplace
 - 2. The positioning of current brands on these dimensions
 - 3. The positioning of consumers ideal brand on these dimensions

Information by MDS has been used for a variety of marketing applications, including:

- Image measurement. Compare the customers' and noncustomers' perceptions of the firm with the firm's perceptions of itself and thus identify perceptual gaps.
- Market segmentation. Position brands and consumers in the same space and thus identify groups of consumers with relatively homogeneous perceptions.
- New product development. To look for gaps in the spatial map, which indicate potential opportunities for positioning new products. Also, to evaluate new product concepts and existing brands on attest basis to determine how consumers perceive the new concepts. The proportion of preferences for each new product is one indicator of its success.
- Assessing advertising effectiveness. Spatial maps can be used to determine whether advertising has been successful in achieving the desired brand positioning.
- Pricing analysis. Spatial maps developed with and without pricing information can be compared to determine the impact of pricing.
- Channel decisions. Judgments on compatibility of brands with different retail outlets could lead to spatial maps useful for making channel decisions.
- Attitude scale construction. MDS techniques can be used to develop the appropriate dimensionality and configuration of the attitude space.



THANK YOU



MODULE-5

DATA ANALYSIS AND REPORT WRITING:

DATA ANALYSIS AND REPORT WRITING: EDITING, CODING, CLASSIFICATION, TABULATION, VALIDATION.

ANALYSIS AND INTERPRETATION, REPORT WRITING AND PRESENTATION OF RESULTS, IMPORTANCE OF REPORT WRITING, TYPES OF RESEARCH REPORTS, REPORT STRUCTURE, GUIDELINES FOR EFFECTIVE DOCUMENTATION

DATA ANALYSIS AND REPORT WRITING: EDITING

- The data which has been collected from various primary/secondary sources is RAW in nature, this means that there are likely chances of errors and inconsistencies in it.
- Since data collected is of pivotal importance to policy and decision makers everywhere be it governmental departments, business organizations, health or educational institutions etc it would be better to have a team of experts at hand who know how to scrutinize, review and **edit** this data before it is finally fed into the data bases and the required statistics are generated.

HOW TO DEFINE DATA EDITING

The process through which the data is reviewed to check for consistency, adequacy, detect errors and outliers (values that are either too big or too small from the rest of the data) and the correction of errors within the data in order to maximize its usefulness for the purpose for which it was collected is called **data editing**.

PURPOSES AND OBJECTIVES OF DATA EDITING

- The basic purpose served by data editing is that it improves the quality, accuracy and adequacy of the collected data thereby making it more suitable for the purpose for which the data was collected. The following can therefore be identified as the main objectives of the data editing process :
- Detection of errors in the data that otherwise affect the validity of outputs.
- Validation of data for the purposes it was collected.
- Provision of information that that would help assess the overall level of accuracy of the data.
- Detection and identification of any inconsistencies in the data and outliers and to make adjustments for them.

TYPES OF DATA EDITING

- **Validity and completeness of data:** refers to correctness and completeness of obtained responses. This helps ensure that there are no missing values or empty fields in the data bases.
- **Range:** verifies that data within a field fall between the boundaries specified for the particular field.
- **Duplicate data entry:** this helps ensure that there is no repetition or duplication of data and each unit on the data base or register was filled only once.
- **Logical consistency:** through this type of editing connections between data fields or variables are taken into account.
- **Outliers:** this type of editing helps detect values that are too extreme or unusual so that they can be verified and checked.

STAGES OF DATA EDITING

- The **manual desk editing** stage is a traditional method that is put into effect by a specialized editing team. The data, (if) on paper is checked after the data has been collected and before it is fed into the data bases. If however, electronic means have been used to collect the data, the forms entered into the database are revised individually.
- The **automated data editing** method makes use of computer programs and systems for checking the data all at once after it has been entered electronically. These programs and systems contain **Audit rules** which validate the data, detect errors and determine unacceptable responses.

LIMITATIONS TO DATA EDITING

- Data editing can be influenced by the amount of time available, the budget, the presence or absence of other resources and also by the group of people involved in the editing process.
- The available computer software programs.
- Follow up with the respondents is of critical importance in the data editing process because they are often the best source of information in many cases. However, the respondents might feel this to be stressful and burdensome thereby causing limitations to the data editing process.
- Some types of data do not require extensive editing, therefore it would be better to keep in mind the intended uses of data and make sure that the more important part of data is kept free from all errors. In this way, the intended use of data does play an important role in influencing the data editing process.
- What you need to do is to establish the methods and procedures that must be followed while correcting or handling the data errors, in the survey plan, right at the start of the project otherwise the process would be of no or little use.
- Also remember that if you plan to edit your data manually, you must develop and document the methods that are to be followed. Your team must be trained, a method must be established to check their work progress and the impact of the edits on the original data must also be assessed.
- In case of automated editing, you would need to develop and document the rules for editing. You might also need to develop a software or customize an existing computer program as per your data editing demands or requirements.

GENERAL GUIDELINES FOR DATA EDITING

- Who should make or set the editing rules? The answer to this question would be that such rules should be made by professionals who are experts in data collection, questionnaire design and analysis.
- The editing rules need to be consistent and free from any contradictions.
- When setting the editing rules, it must be established whether the variable is qualitative or quantitative because the rules for editing either one are different from the other.
- Give enough time to each of the various stages of the process, that is, data collection entry and analysis and at the end of each make a quick check to see that all the necessary edits have been made and that there are no empty places within the questionnaire form.
- The questionnaire must be edited in full during the early stages of editing. If however it is found that some errors remain, a sample of forms should be subjected to re editing. The size of the sample is determined according to the expected number of the remaining errors.
- You also need to re run the desk editing stage to ensure that the data is almost free from all errors.
- The questionnaire must be subjected to desk editing and also to automated rules within the built in data editing/computer software programs.

CODING

When a researcher has completed collecting information or data, this information is ready to be processed and analyzed. Quantitative data is information that is measurable and focuses on numerical values, unlike qualitative data which is more descriptive. During the data processing step, the collected data is transformed into a form that is appropriate to manipulate and analyze.

The process in which raw data is transformed into a standardized form suitable for machine processing and analysis is called coding. In other words, coding is the act of assigning numerical values to a set of data in order to make the analysis simpler. Coding can be used to quantify both manifest content i.e. the tangible or concrete surface content (data), and latent content i.e. the underlying meaning behind this information. The difference between manifest content and latent content is very important when it comes to survey research.

IMPORTANT TO KEEP THE FOLLOWING POINTS IN MIND DURING CODING OF DATA:

- Identification variables:-
- Code categories :-
- Preserving original information:-
- Closed-ended questions:-
- Open-ended questions:-
- Check-coding:-
- Series of responses:-

DEALING WITH MISSING DATA

- Refusal to answer or No response
- Don't know responses
- Processing error
- Not applicable
- No match
- No data available

CLASSIFICATION

- **Meaning of Classification of Data**

- It is the process of arranging data into homogeneous (similar) groups according to their common characteristics.
- Raw data cannot be easily understood, and it is not fit for further analysis and interpretation. Arrangement of data helps users in comparison and analysis.
- For example, the population of a town can be grouped according to sex, age, marital status, etc.
- **Definition of classification given by Professor. Secrist** - “Classification is the process of arranging data into sequences according to their common characteristics or separating them into different related parts.”

OBJECTIVES OF DATA CLASSIFICATION

- To consolidate the volume of data in such a way that similarities and differences can be quickly understood. Figures can consequently be ordered in sections with common traits.
- To aid comparison.
- To point out the important characteristics of the data at a flash.
- To give importance to the prominent data collected while separating the optional elements.
- To allow a statistical method of the materials gathered.

TABULATION

Tabulation is the systematic and logical representation of figures in rows and columns to ease comparison and statistical analysis. It eases comparison by bringing related information closer to each other and helps further in statistical research and interpretation. In other words, tabulation is a method of arranging or organizing data in a tabular form. The tabulation process may be simple or complex depending upon the type of categorization.

WHAT ARE THE ESSENTIAL PARTS OF A TABLE?

- Table Number –
- Title of the Table –
- Headnote –
- Column Headings or Captions –
- Row Headings or Stubs -
- Body of a Table –
- Footnote –
- Source Note –

TYPES OF TABULATION

- Simple Tabulation or One-way Tabulation
- Double Tabulation or Two-way Tabulation
- Complex Tabulation

WHAT ARE THE OBJECTIVES OF TABULATION?

- **For Simplification of Complex Data –**
- **To Highlight Important Information –**
- **To Enable Easy Comparison –**
- **To Help in the Statistical Analysis of Data –**
- **Saves Space**

VALIDATION

- **What is Data Validation?**
- Data validation refers to the process of ensuring the accuracy and quality of data. It is implemented by building several checks into a system or report to ensure the logical consistency of input and stored data.
- In automated systems, data is entered with minimal or no human supervision. Therefore, it is necessary to ensure that the data that enters the system is correct and meets the desired quality standards. The data will be of little use if it is not entered properly and can create bigger downstream reporting issues. Unstructured data, even if entered correctly, will incur related costs for cleaning, transforming, and storage.

TYPES OF DATA VALIDATION

1. Data Type Check
2. Code Check
3. Range Check
4. Format Check
5. Consistency Check
6. *Uniqueness Check*

ANALYSIS AND INTERPRETATION

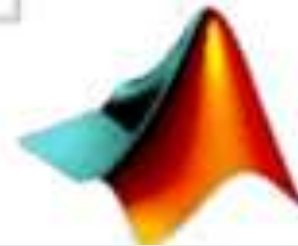
Data analysis is defined as a process of cleaning, transforming, and modeling data to discover useful information for business decision-making. The purpose of Data Analysis is to extract useful information from data and taking the decision based upon the data analysis.

A simple example of Data analysis is whenever we take any decision in our day-to-day life is by thinking about what happened last time or what will happen by choosing that particular decision. This is nothing but analyzing our past or future and making decisions based on it. For that, we gather memories of our past or dreams of our future. So that is nothing but data analysis. Now same thing analyst does for business purposes, is called Data Analysis.

DATA ANALYSIS TOOLS



SQL



MATLAB

TYPES OF DATA ANALYSIS: TECHNIQUES AND METHODS

- Text Analysis
- Statistical Analysis
- Diagnostic Analysis
- Predictive Analysis
- Prescriptive Analysis

DATA ANALYSIS PROCESS

- The **Data Analysis Process** is nothing but gathering information by using a proper application or tool which allows you to explore the data and find a pattern in it. Based on that information and data, you can make decisions, or you can get ultimate conclusions.
- Data Analysis consists of the following phases:
- Data Requirement Gathering
- Data Collection
- Data Cleaning
- Data Analysis
- Data Interpretation
- Data Visualization

DATA INTERPRETATION

- After analyzing your data, it's finally time to interpret your results. You can choose the way to express or communicate your data analysis either you can use simply in words or maybe a table or chart. Then use the results of your data analysis process to decide your best course of action.
- **Data Visualization**
- Data visualization is very common in your day to day life; they often appear in the form of charts and graphs. In other words, data shown graphically so that it will be easier for the human brain to understand and process it. Data visualization often used to discover unknown facts and trends. By observing relationships and comparing datasets, you can find a way to find out meaningful information.

IMPORTANCE OF DATA INTERPRETATION

- **Make better decisions**
- **Find trends and take action**
- **Better resource allocation**

WHAT ARE THE STEPS IN INTERPRETING DATA?

- Gather the data
- Develop your discoveries
- Draw Conclusions
- Give recommendations

REPORT WRITING AND PRESENTATION OF RESULTS,

- **Report writing**
- Reports are usually spread across a vast horizon of topics but are focused on communicating information about a particular topic and a niche target market. The primary motive of research reports is to convey integral details about a study for marketers to consider while designing new strategies.
- Certain events, facts, and other information based on incidents need to be relayed to the people in charge, and creating research reports is the most effective communication tool. Ideal research reports are extremely accurate in the offered information with a clear objective and conclusion. These reports should have a clean and structured format to relay information effectively.

WHAT ARE RESEARCH REPORTS?

- Research reports are recorded data prepared by researchers or statisticians after analyzing the information gathered by conducting organized research, typically in the form of surveys or qualitative methods.
- A research report is a reliable source to recount details about a conducted research. It is most often considered to be a true testimony of all the work done to garner specificities of research.
- The various sections of a research report are:
 - Summary
 - Background/Introduction
 - Implemented Methods
 - Results based on Analysis
 - Deliberation
 - Conclusion

COMPONENTS OF RESEARCH REPORTS

- Components of Research Reports
- Research Report Summary:
- Research Introduction:
- Research Methodology:
- Research Results:
- Research Discussion:
- Research References and Conclusion

TIPS FOR WRITING RESEARCH REPORTS

1. Prepare the context before starting to write and start from the basics:
2. Keep the target audience in mind while selecting a format that is clear, logical and obvious to them:
3. Have a clear research objective:
4. Establish a working model:
5. Gather all the information about the research topic.
6. Read aloud while writing.
7. Check grammar and spelling.
8. Discuss only the discoveries that are significant.
9. Try and stick to the survey questions.
10. The graphs must be clear enough so that they understand themselves.
11. Be clear with messages.
12. Be creative with titles
13. Create an effective conclusion
14. Get a couple more pair of eyes to read the report.

PRESENTATION OF RESULTS

- **Written Presentation of Results**
- Once you've gone through the process of doing communication research – using a quantitative, qualitative, or critical/rhetorical methodological approach – the final step is to communicate it.
- The major style manuals (the APA Manual, the MLA Handbook, and Turabian) are very helpful in documenting the structure of writing a study, and are highly recommended for consultation. But, no matter what style manual you may use, there are some common elements to the structure of an academic communication research paper.

STRUCTURE OF PROJECTS

Social Scientific Project

- Introduction
- Literature Review
- Method
- Results
- Discussion
- References

Critical Projects

- Introduction
- Definition of Terms
- Literature Review
- Analysis (divide sections by areas of investigation)
- Conclusions*

*All sections may not be named this way

IMPORTANCE OF REPORT WRITING

1. Decision Making Tool:
2. Investigation:
3. Evaluation:
4. Quick Location:
5. Development of Skill:
6. Neutral Presentation of Facts:
7. Professional Advancement:
- Proper Control:
9. A Managerial Tool:
10. Encountering Advance and Complex Situation:

TYPES OF RESEARCH REPORTS

1. Thesis
2. Research Paper
3. Technical Report
4. Progress Report
5. Feasibility Report
6. Field Report
7. Experimental Report
8. Case Study Report
9. Literature Review Report

REPORT STRUCTURE

- **(A)-Front Matter**
 - 1) Letter of transmittal (Cover letter)
 - 2) Title page
 - 3) Executive summary
 - 4) Table of content
 - 5) List of figures/ Illustration
- **(B)-Body of the report**
 - 6) Introduction
 - 7) Findings and Documentation
 - 8) Conclusion and Recommendation
- **(C)-Back Matter**
 - 9) Appendices, References, and Glossary
- Note: The letter of transmittal is an optional element of reporting which may or may not be included depending on the type of report prepared.

GUIDELINES FOR EFFECTIVE DOCUMENTATION

Rule 1 Never consider the time you invest in writing or planning documentation as wasted time

Rule 2. Don't take what you say or write for granted

Rule 3. Provide plenty of feedback and input

Rule 4. Be prepared for several drafts

Rule 5. Be on friendly terms with your writer

Rule 6. Make sure all stages of design are well-documented

Rule 7. Involve your writer in all stages of the product design

Rule 8. Writing is team work

Rule 9. Be Flexible in your expectations and demands

THANK YOU

“No data is clean,
but most is useful.”

Dean Abbott

Co-founder and Chief Data Scientist at SmarterHQ



Module-6

INTELLECTUAL PROPERTY RIGHTS:

Syllabus

Intellectual Property Rights: Meaning and Concepts of Intellectual Property, Nature and Characteristics of Intellectual Property, Origin and Development of Intellectual Property, Kinds of Intellectual Property, Intellectual Property System in India, IPRs- Invention and Creativity- Intellectual Property-Importance and Protection of Intellectual Property Rights (IPRs)- **A brief summary of:** Patents, Copyrights, Trademarks, TRIPS and TRIMS , Industrial Designs- Integrated Circuits-Geographical Indications-Establishment of WIPO- Application and Procedures.

What is the Definition of Intellectual Property Rights?

- ▶ The definition of intellectual property rights is any and all rights associated with intangible assets owned by a person or company and protected against use without consent. Intangible assets refer to non-physical property, including right of ownership in intellectual property . Examples of intellectual property rights include:
 - ▶ Patents
 - ▶ Domain names
 - ▶ Industrial design
 - ▶ Confidential information
 - ▶ Moral rights
 - ▶ Database rights
 - ▶ Works of authorship
 - ▶ Service marks
 - ▶ Logos
 - ▶ Trademarks
 - ▶ Design rights
 - ▶ Business or trade names
 - ▶ Commercial secrets
 - ▶ Computer software

What Are the Types of Intellectual Property?

There are four main types of intellectual property rights, including patents, trademarks, copyrights, and trade secrets. Owners of intellectual property frequently use more than one of these types of intellectual property law to protect the same intangible assets. For instance, trademark law protects a product's name, whereas copyright law covers its tagline.

1. Patents

The U.S. Patent and Trademark Office grants property rights to original inventions, from processes to machines. Patent law protects inventions from use by others and gives exclusive rights to one or more inventors. Technology companies commonly use patents, as seen in the [patent for the first computer](#) to protect their investment in creating new and innovative products. The three types of patents consist of:

Design patents: Protection for the aesthetics of a device or invention. Ornamental design patents include a product's shape (Coca-Cola bottle), emojis, fonts, or any other distinct visual traits.

Plant patents: Safeguards for new varieties of plants. An example of a plant patent is pest-free versions of fruit trees. But inventors may also want a design patent if the tree has unique visual properties.

Utility patents: Protection for a product that serves a practical purpose and is useful. IP examples include vehicle safety systems, software, and pharmaceuticals. This was the first, and is still the largest, area of patent law.

2. Trademarks

Trademarks protect logos, sounds, words, colors, or symbols used by a company to distinguish its service or product. Trademark examples include the Twitter logo, McDonald's golden arches, and the font used by Dunkin'.

Although patents protect one product, trademarks may cover a group of products. The Lanham Act, also called the Trademark Act of 1946, governs trademarks, infringement, and service marks.

3. Copyrights

Copyright law protects the rights of the original creator of original works of intellectual property. Unlike patents, copyrights must be tangible. For instance, you can't copyright an idea. But you can write down an original speech, poem, or song and get a copyright.

Once someone creates an original work of authorship (OWA), the author automatically owns the copyright. But, registering with the [U.S. Copyright Office](#) gives owners a head-start in the legal system.

4. Trade Secrets

Trade secrets are a company's intellectual property that isn't public, has economic value, and carries information. They may be a formula, recipe, or process used to gain a competitive advantage.

To qualify as a trade secret, companies must work to protect proprietary information actively. Once the information is public knowledge, then it's no longer protected under trade secrets laws. According to [18 USC § 1839\(3\)](#), assets may be tangible or intangible, and a trade secret can involve information that's:

Business

Financial

Technical

Economic

Scientific

Engineering

Two well-known examples include the recipe for Coca-Cola and Google's search algorithm. Although a patent is public, trade secrets remain unavailable to anyone but the owner.

What Are Some Examples of Violations of Intellectual Property?

The significant violations of intellectual property consist of infringement, counterfeiting, and misappropriation of trade secrets. Violations of intellectual property include:

Creating a logo or name meant to confuse buyers into thinking they're buying the original brand

Recording video or music without authorization or copying copyrighted materials (yes, even on a photocopier, for private use)

Copying another person's patent and marketing it as a new patent

Manufacturing patented goods without a license to do so

Since intellectual property can be bought, sold, or leased out, it offers many protections equal to real property ownership. Likewise, similar remedies exist. A dispute may end with property confiscation, an order of monetary damages, or cease and desist orders.

Concept of Intellectual Property Rights

Intellectual property rights are the rights given to persons over the creations of their minds, such as inventions, literary and artistic works, designs, and symbols, names, and images used in commerce. Intellectual property is the product of human intellect including creativity, concepts, inventions, industrial models, trademarks, songs, literature, symbols, names, brands, etc.

Nature & characteristics of Intellectual Property Right:

- (1) Intangible property:
- (2) Rights & duties
- (3) creation of statute:
- (4) Territoriality
- (5) Assignable
- (6) Dynamism
- (7) subject to public policy
- (8) subject matter of IPR protection

Origin of Intellectual Property (IP)

- ▶ The Intellectual Property (IP) is a term referred to work or inventions that are a result of some individual's original creativity. It basically, means the legal rights arising out of an intellectual activity in any field like literary, industrial and artistic or industrial etc. It includes the results that are the creations of one's creative mind. Prior to the General Agreement on Tariffs and Trade (GATT), the Intellectual Property and its related rights were not a subject, to any international trade negotiations. There's a long history of the Intellectual Property, which is way complex but, also fascinating. It is traced all the way to 500 BCE, when Sybaris, a Greek State made it possible for the citizens of their state, to obtain a patent for one year, for "any new refinement in luxury". Since, then we can conclude that Patent, Copyright and Trademark laws have become more complicated over the centuries but, the intent remains the same.
- ▶ The laws and legislation procedures relating to the IPR have their roots in Europe. The trend of Patents started in the 14th Century, they were technologically less advanced than England. However, the first ever known Copyrights appeared to be in Italy where, Venice was considered the cradle of Intellectual Property systems. While, the Patents are about 150 years old concept, as first introduction was based on, the British Patent System. To foster creativity and to ensure the possibility for the inventor to make benefits of their creativity.

Development of Intellectual Property

- ▶ With the ongoing development around it was necessary to protect the rights of the owner of intellectual property since without protecting the innovative work that a person has spent his time on, there would be no promotion of invention and creativeness and without such development it would be impossible to expect the market grow beyond a close boundary.
- ▶ Development can only be persisted if the rights of the producers and the one who are sweating over the invention are getting protection for their hard work. Without providing exclusivity to them, the market would copy each other and there would be no mark of quality and would be tough for the sellers as well as the buyers in the market to choose product of a quality.

Kinds of Intellectual Property

Copyright

- ▶ The term ‘copyright’ concerns the rights of the creators/authors of literary and artistic works. A copyright is also called a ‘literary right’ or ‘author’s right’. Copyright gives an author exclusive rights to his creation and prevents the copying and unauthorised publishing of his work. Copyright protection begins at the very moment a work is created and expressed in some tangible form. Copyright protection is granted to a work that is an original creation. Also, the protection extends only to expressions. Mere ideas without any tangible expression are not granted legal protection and do not form the subject matter of copyright. Copyright protects the following two rights of the author:
- ▶ Economic rights i.e., the right of the owner to derive financial benefit from the use of their works by others. For instance, the right to prohibit or authorise reproduction of the work in various forms, the right to prohibit unauthorised translation of the work, etc.
- ▶ Moral rights i.e., protection of non-economic interests of the author. For instance, the right to oppose changes to work and the right to claim authorship, etc.

Patents

- ▶ A patent is an exclusive right granted for an invention or innovation, which might be a product, a method or a process, that introduces a novel way of doing something or offers a new technical solution to a problem. In other words, it is a right of monopoly granted to a person who has invented:
- ▶ a new and useful article, or
- ▶ improvement of an existing article, or
- ▶ a new process of making an article.
- ▶ A patent is granted for inventions having industrial and commercial value. It is the exclusive right to manufacture the new article/manufacture the article with the invented process for a limited period of time (usually 20 years from the filing date of the application) in exchange for disclosure of the invention. A patent owner can sell his patent or grant licence to others to exploit the same.

Types of Patents

- ▶ **Utility Patents**
- ▶ **Design Patents**
- ▶ **Plant Patents**

How to Apply for a Patent

- ▶ Before making a formal application, an applicant should research the Patent and Trademark Office's database to see if another person or institution has claimed a patent for a similar invention. The invention must be different from or an improvement upon a previous design to be considered for a patent. Applicants need to take care to maintain accurate records of the design process and the steps taken to create the invention. Enforcing the patent is up to the person or entity that applied for the patent.
- ▶ To apply for a patent in the United States, the applicant submits specific documents and pays associated fees. Written documentation includes drawings, descriptions, and claims of the item to be patented. A formal oath or declaration confirming the authenticity of the invention or improvement of an existing invention must be signed and submitted by the inventor. After fee payment, the application is reviewed and either approved or denied.

Examples of Patents

- ▶ One of the most notable patents in the past 40 years was the personal computer filed in 1980 by Steve Jobs and three other employees of Apple Inc.⁵
- ▶ King C. Gillette patented the razor in 1904 and was dubbed a "safety razor."⁶ Garrett Morgan was granted a patent for the traffic light in 1923.⁷ The patent for the television was issued in 1930 to Philo Taylor Farnsworth for the "first television system."⁸
- ▶ United States Patent and Trademark Office. "[Patent 1,773,980.](#)"
- ▶ At age 20, Farnsworth had created the first electric television image and went on to invent an early model of the electronic microscope.

Trademarks and service marks

- ▶ TA trademark is a symbol that is used to distinguish the goods of one enterprise from its competitors. A trademark may consist of a single letter, logo, symbol, design, or numerals and three-dimensional features such as shape and packaging, etc. Section 2(zb) of the Trademarks Act, 1999 defines “trademark” as a mark capable of graphical representation and which can be used to distinguish the goods or services of one person from those of others. A trademark may include the shape of goods, their packaging, and a combination of colours. Hence, distinctiveness is the hallmark of a trademark.
- ▶ Trademarks used in connection with services such as tourism, banking, etc., are called Service Marks.

Procedure for Registration of Trade Mark

- ▶ Any mark which shows just kind, quality, quantity or intended purpose of any goods, e.g. trademark of any goods can't be “500 gram”
- ▶ Mark consisting of marks or indications which are customary in current language, e.g. OMG (abbreviation of Oh My God) cannot be trade mark.
- ▶ However where trademark which bears distinctive character because of its use, or is well-known to public is allowed, e.g. “Sugar-free” mark shows just quality but still allowed because this product is well-known to public
- ▶ If a mark is capable to deceive or cause confusion to public, or hurt religious susceptibility, or is obscene, it will not be registered — Mark consisting of shape of natural goods (e.g. Neem leaf) or such shape which is necessary to obtain a given technical result shall not be registered.

How Copyrighting Works

- ▶ When someone creates a product that is viewed as original and that required significant mental activity to create, this product becomes an intellectual property that must be protected from unauthorized duplication. Examples of unique creations include:
- ▶ Novels
- ▶ Art
- ▶ Poetry
- ▶ Musical lyrics and compositions
- ▶ Computer software
- ▶ Graphic designs,
- ▶ Film
- ▶ Original architectural designs
- ▶ Website content

Geographical Indications (GI)

- ▶ A geographical indication (GI) is used to identify goods having a specific geographical origin. These indications denote quality, reputation, or other characteristics of such goods essentially attributable to their geographical origin. Generally, geographical indications are used for foodstuffs, agricultural products, wine, industrial products and handicrafts. Examples of GI include Basmati Rice, Darjeeling Tea etc.

Trade Secrets

- ▶ Trade Secrets are IP rights on confidential information which may be sold or licensed. A trade secret refers to any confidential business information and may include designs, drawings, plans, business strategies, R & D related information, etc. In order to qualify as a trade secret, the information should be commercially valuable i.e. useful in a trade or business, known to a small number of people, and subject to reasonable steps taken by the rightful holder of the information to keep it secret.

Layout designs of integrated circuits

- ▶ Integrated circuits are used in products such as television, radio, mobile, washing machine, and data processing instruments. The layout designs of integrated circuits not only reduce the space but also enhance the capacity and performance of the system. In India, the Semiconductor Integrated Circuit Layout Design Act, 2000 regulates the registration, use, and protection of original and distinct layout designs.

The Intellectual Property System in India

- ▶ The origins of India's IPR system date back to British colonial rule, when as a colony the state enacted various rules and enforcement mechanisms pertaining to IP rights. Post-independence, India retained elements of these structures while updating some guiding regulations and other bureaucratic structures. As India moved toward liberalization, privatization, and globalization in the 1990s and later, Indian policymakers made further adjustments to keep up with growing needs of domestic and international stakeholders. Indian IPR Laws fully conform to the Agreement on Trade Related Aspects of Intellectual Property Rights under WTO aegis.

The objectives of the Policy are:

1. **IPR Awareness: Outreach and Promotion** - To create public awareness about the economic, social and cultural benefits of IPRs among all sections.
 2. **Generation of IPRs - To stimulate the generation of IPRs:** India has a large talent pool of scientific and technological talent spread over R&D institutions, enterprises, universities and technical institutes. There is a need to tap this fertile knowledge resource and stimulate the creation of IP assets. (see **Annexures II and III** for no. of applications for patents and top patentees.)
 3. **Legal and Legislative Framework** - To have strong and effective IPR laws, which balance the interests of rights owners with larger public interest.
 4. **Administration and Management** - To modernize and strengthen service- oriented IPR administration.
 5. **Commercialization of IPR** - Get value for IPRs through commercialization.
 6. **Enforcement and Adjudication** - To strengthen the enforcement and adjudicatory mechanisms for combating IPR infringements.
 7. **Human Capital Development** - To strengthen and expand human resources, institutions and capacities for teaching, training, research and skill building in IPRs.
- The present IPR Policy aims to integrate IPR as a policy and strategic tool in national development plans. It foresees a coordinated and integrated development of IP system in India and the need for a holistic approach to be taken on IP legal, administrative, institutional and enforcement related matters.

IPRs- Invention and Creativity

What is innovation?

- ▶ So what does **innovation** really mean? And how does it differ from **invention** and **creativity** in a business? Let's start from a definition of innovation, invention and creativity:
- ▶ **Innovation** is the process of turning a new concept into commercial success or widespread use.
- ▶ **Invention** is the creation of a new idea or concept.
- ▶ **Creativity** is the act of turning new and imaginative ideas into reality.
- ▶ **Creativity, invention, and innovation** are all interrelated and necessary for growth to occur. We can follow any successful company and see an investment of time and effort into these three concepts. Based on our experience and research conducted, we have discovered these insights:

Creativity

- ▶ **Creativity is the act of channeling imagination into something concrete.** It is the very first stage of design, where ideas start to actually take form, and a plan can be developed.
- ▶ **Examples of creativity:**
- ▶ in the case of Uber, creativity was necessary in producing the **concept of ride sharing** and of making driving jobs accessible to any individual.
- ▶ another example can be the **Google Glass**, which although were very creative at the time (new, different in the market), were not necessarily innovative (monetary return over investment made by the company).

Importance of IPR

Positive Impacts :

- ▶ IPRs play very important role in the progress and development of the society. IPRs not only provide incentive to the creator of his creation but also lead to a healthy competition among creators which ultimately leads to the progress of the society. Some of the positive impacts of IPR are:
- ▶ IPRs are important for free flow for energy for enhancing invention and research. IPRs provide incentive to the individuals for new creations. IPRs provide due recognition to the creators and inventors. These laws provide them both the means and incentive to create newer works, products and services
- ▶ Intellectual Property Rights enhance innovation and creativity by protecting the rights of inventors and artists. Since the filing of patents requires the disclosure of information that would enable others to replicate the inventor's discovery, others can use and build upon this shared knowledge to create newer and/or better products. IPRs ensure material reward for intellectual property.
- ▶ IPRs ensure the availability of the genuine and original products. Intellectual Property rights, such as patent and copyrights, are an important means used by firms to help protect their investments in innovation (Shankar Narayanan, 2010).
- ▶ IPRs may be helpful in the solution to global challenges like in the field of alternate sources of energy, new products to the farmers and development of low cost drugs for poor people.
- ▶ IPRs are necessary to stimulate economic growth. Protection of intellectual property rights is essential in maintaining economic growth. They encourage fair trading which would contribute to economic and social development. Effective enforcement of intellectual property rights is critical to sustaining economic growth across all industries and globally.

Negative Impacts:

- ▶ Though the purpose of IPRs is to enhance innovation and creativity by protecting the rights of creators and leading to the growth and advancement of human beings but do IPRs actually
- ▶ play a constructive role in the progress and development of society and mankind or it is a form of intellectual protectionism or a form of a temporary monopoly enforced by the state, is an issue of great concern.
- ▶ They are considered to benefitting concentrated i.e. IPRs confer authority over resources to a few. The few gain power over the goals of many. The objective of IPRs is to protect the public interest but, in fact, the public interest is harmed. Pharmaceutical product prices form substantial portion of health care costs and strong intellectual property protection is one of the major reasons for high health care costs (Agatha, 2013). For example, patent of life saving drugs had allowed the countries to charge higher than the marginal cost of the production in the name of cost of research and development. This has led to an increase in the cost of drugs and they are not affordable by the poor sections of the society. When there is control by some particular group, IPRs actually discourage invention. Patent of the ideas may prevent the owner of a property to utilize it according to his wishes. In order to serve the purpose IPRs must focus on the needs of the poor and developing countries but in reality intellectual properties tend to be governed by economic goals preventing the progress of poor. Thus, IPRs are considered as intellectual protectionism, intellectual monopoly or government- granted monopoly by which the public interest is harmed and progress is stopped.

What is TRIPS?

- ▶ The Trade-Related Aspects of Intellectual Property Rights (TRIPS) adheres to member states' core trade principles concerning intellectual property; all WTO members must abide by it. The TRIPS Agreement stipulates reasonable exceptions and constraints to reconcile intellectual property rights with community health and economic progress objectives.
- ▶ The most robust international IP agreement, TRIPS, tends to play an influential role in fostering commerce in creativity and knowledge, addressing trade disputes comprising intellectual property, and enabling WTO members with the autonomy to pursue their domestic policy goals. The agreement also shapes the intellectual property system regarding innovations, technology transmission, and public welfare.
- ▶ The TRIPS Agreement must be assessed and implemented under the jurisdiction of the TRIPS Council. The TRIPS agreement was outlined at the time of the Uruguay Round that lasted from 1986 to 1994 for GATT. The terminology "Berne and Paris-plus" also refers to the TRIPS Agreement.

Significance of TRIPS

- ▶ The TRIPS Agreement has its key focus on safeguarding intellectual properties in the World Trade Organization and the international trading system. The TRIPS agreement is often a pillar of the World Trade Organization. The remaining two are trading in services and commodities (the traditional competence of the GATT).
- ▶ Before TRIPS, there were notable sectoral discrepancies in the range of IP protection and enforcement. However, as IP grew more crucial to commerce, these discrepancies became a focus of contention in international economic relations. To have more order and reliability and to adjudicate controversies in an orderly manner, it was therefore thought advisable to endorse new trade regulatory requirements for IP rights.

TRIMS

- ▶ The agreement on Trade-Related Investment Measures or “TRIMs” acknowledges that investment measures can control and twist trade. This states World Trade Organisation members might not apply any measure that differentiates or leads to quantitative restrictions, both of which disobey any fundamental WTO rules.
- ▶ A list of restricted TRIMS is a part of this agreement. The TRIMS Committee looks after the operation and execution of this agreement and allows members the chance to consult on any appropriate matters.
- ▶ Agreement on TRIMs resulting from Uruguay recognises that certain investment measures may cause restrictive effects on international trade in goods.

Objectives of Trade-Related Investment Measures

- ▶ TRIMs believe that there is a strong connection between trade and investment. The goal of trade-related investments measures is to give fair treatment to all investing members across the world.
- ▶ As the TRIMs deal says, members have to inform the World Trade Organization (WTO) council to buy and sell various services and goods of their current TRIMs that are incompatible with the agreement.

Main Features of TRIMs

- ▶ It only applies to investment measures related to goods trade.
- ▶ This doesn't apply to service trade.
- ▶ It doesn't regulate the entry of foreign industry or investment.
- ▶ It is about the discriminatory treatment of imported/exported products.
- ▶ Concern measures were applied to both foreign domestic firms.
- ▶ A transition period of 2 years in the case of developed countries, 5 years in the case of developing countries and 7 years in the case of LDCs, from the date this agreement came into effect, which is 1st January 1995.

Industrial Design

- ▶ The owner of a design patent or an industrial design that's registered has the rights to stop third parties from the following:
- ▶ Selling articles that have a copy of the design
- ▶ Making articles with a similar design
- ▶ Importing articles that bear a design that's a copy of the original one
- ▶ Copies of any form are not allowed when the purpose is for commercial use. There is a wide range of products of handicraft and industry items that these designs apply to. This includes containers, packages, household goods, furnishing, jewelry, lighting equipment, textiles, and electronic devices. Industrial designs also apply to graphical user interfaces, logos, and graphic symbols. Most countries require an industrial design to be registered so it can be protected under industrial design and considered a registered design. Some countries allow unregistered industrial designs and scope limited protection.

What is an Integrated Circuit (IC)?

- ▶ Before the discovery of ICs, the basic method of making circuits was to select the components like diodes, transistors, resistors, inductors and capacitors and connect them by soldering. But due to size and power consumption issues, it was necessary to develop a small size circuit with less power consumption, reliability and shockproof.
- ▶ After the invention of the semiconductors and transistors, things were quite simplified to a particular extent, but the development of integrated circuits changed electronics technology's face. Jack Kilby from Texas Instruments and Bob Noyce from Intel are the official creators of integrated circuits, and they did it independently.
- ▶ The integrated circuit is a fundamental concept of electronics that builds on other basic concepts previously discussed in our syllabus. Therefore, for a quick reference, go through the articles listed below:

- ▶ [Electric Circuits](#)
- ▶ [Resistors](#)
- ▶ [Transistors](#)
- ▶ [Diodes](#)
- ▶ [Capacitors](#)

INTEGRATED CIRCUIT



Integrated Circuit Design

Digital Design

- ▶ The digital design approach is used to create integrated circuits (ICs), which are utilised as computer memories (such as RAM and ROM) and microprocessors. With this approach to design, the circuit density and overall efficiency are both maximised. The ICs created with this technique operate with binary input data like 0 and 1. The process for designing digital integrated circuits is depicted in the diagram below.

Analog Design

- ▶ IC chip is created by using the analogue design process when:
- ▶ ICs are utilised as regulators, filters and oscillators.
- ▶ Optimal power dissipation, gain and resistance are required.

Mixed Design

- ▶ The analog and digital design ideas are used in mixed designs. The mixed ICs perform either Analog to Digital or Digital to Analog conversions.

Establishment of WIPO

► Introduction

- WIPO, which stands for the World Intellectual Property Organisation, was formed on 14th July 1967 and it is headquartered in Geneva, Switzerland. This article will discuss the history of WIPO, how it took the shape into what it is now, what are the missions, functions, goals of WIPO and how it is working for the betterment of Intellectual Property rights around the world.

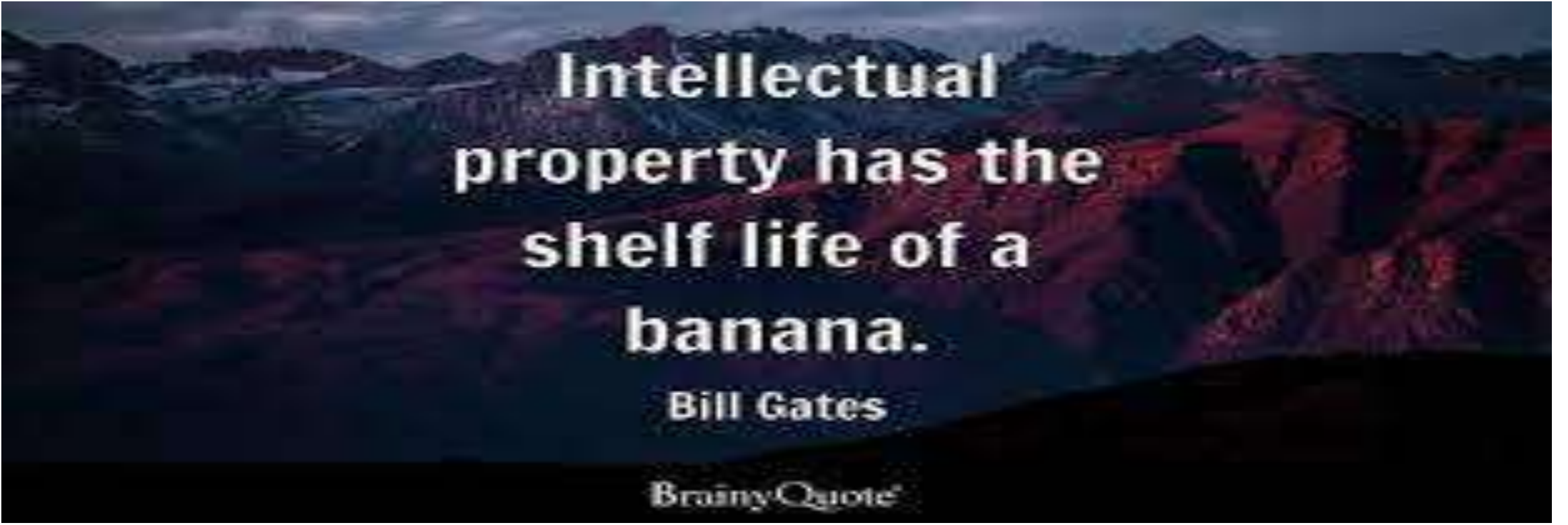
World Intellectual Property Organisation

- ▶ World Intellectual Property Organisation is an organisation based in Geneva and it works with the vision of encouraging creative activity and for promoting the protection of Intellectual Property throughout the world. WIPO is one of the 15 specialised agencies of the United Nations. Currently, there are 193 members in the World Intellectual Property Organisation. WIPO, at the time when it started, was originally about promoting the protection of intellectual property, but when it joined the United Nations in 1974, the objective was redefined as public interest or humanitarian goal. Article 1 of the key agreement establishing WIPO's relationship to the UN restates WIPO's purpose as: "for promoting creative intellectual activity and for facilitating the transfer of technology related to the industrial property to the developing countries in order to accelerate economic, social and cultural development..."
- ▶ WIPO is a unique organisation among the UN organisations, its activities largely are self-funded, according to the Program and Budget of WIPO for the year 2020-2021, the expected income for the biennium will be over 880 million Swiss francs and the expected expenditure is 768 million Swiss francs. WIPO has expected that nearly 95% of the expected income is going to be generated from the fees that the organisation will receive for their services.
- ▶ The World Intellectual Property Organisation is the oldest organization in the field of IP protection. Actually, it was created at the diplomatic conference in 1893.

The main functions of WIPO include:

- ▶ Assisting campaigns development to improve IP protection all over the world and to harmonize national legislation in this field,
- ▶ Signing the international agreements on IP protection,
- ▶ Applying the administrative functions of the Paris and Berne Unions,
- ▶ Rendering technical and legal assistance in the field of IP,
- ▶ collecting and disseminating the information, conducting researches and publishing their results,
- ▶ ensuring the work of the services facilitating the international IP protection,
- ▶ applying any other appropriate actions.

THANK YOU



**Intellectual
property has the
shelf life of a
banana.**

Bill Gates

BrainyQuote

